

UNITROL[®] 1000-15

User's Manual



Automatic Voltage Regulator

Compact voltage regulator for synchronous machines up to 15 A exciter current.

Revision	
Control:	4.3xx
Panel:	4.3xx
CMT1000:	4.3xx

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 ABB Switzerland Ltd			Document number 3BHS201795 E80	Lang. en	Rev. ind. C	Page 1

Important Notice

Our experience has shown that, if the information and recommendations contained in this Operating Instructions are observed, the best possible reliability of our products is assured.

The data contained herein purports solely to describe the product and is not a warranty of performance or characteristics. It is with the best interests of our customers in mind that we constantly strive to improve our products and keep them abreast of advances in technology. This may, however, lead to discrepancies between a product and its "Technical Description" or "Operating Instructions".

This document has been carefully prepared and reviewed, however should in spite of this the reader find an error, he is requested to inform us at his earliest convenience.

It is scarcely possible for the operating instructions for technical equipment to cover every eventuality, which can occur in practice. We would therefore request you to notify us or our agent in the case of all unusual behaviour that does not appear to be covered by these operating instructions.

It is pointed out that all local regulations must be observed when connecting and commissioning this equipment in addition to these operating instructions.

We cannot accept any responsibility for damage incurred as a result of mishandling the equipment regardless of whether particular reference is made in these operating instructions or not.

We lay particular stress on the fact that only genuine spare parts should be used for replacements.

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1 SAFETY INSTRUCTIONS

1.1 General

The safety instructions shall be followed during installation, commissioning, operation and maintenance of the excitation system. Read all instructions carefully before operating the device and keep this manual for future reference.

Required Qualification

Personnel involved in installation work and commissioning of the UNITROL 1000-15 must be familiar, specially instructed and informed about the residual danger areas according to the regulations currently in force.

Operating personnel is not permitted to work at the control system.

Specially instructed personnel must only carry out maintenance and repair work.

The maintenance personnel must be informed about the emergency shutdown measures and must be capable of turning off the system in case of emergency.

The maintenance personnel must be familiar with the accident prevention measures at their workplace and must be instructed in first aid and fire fighting.

Responsibilities

It is the owner's responsibility to ensure that each person involved in the installation and commissioning of the UNITROL 1000-15 has received the appropriate training or instructions and has thoroughly read and clearly understood the safety instructions in this chapter.

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1.2 Safety Instructions

The safety instructions always appear at the beginning of each chapter and/or precede any instruction in the context where a potentially dangerous situation may appear. The safety instructions are divided into five categories and emphasized by the use of the following layout and safety signs:



DANGER!

This symbol indicates an imminent danger resulting from mechanical forces or high voltage. A non-observance leads to life-threatening physical injury or death.



WARNING!

This symbol indicates a dangerous situation. A non-observance may lead to bad or life-threatening physical injury or death.



CAUTION!

This symbol indicates a dangerous situation. A non-observance may lead to physical injury or cause damage to the converter.



NOTICE!

This symbol emphasizes important information. A non-observance may cause damage to the converter or to objects close to it.



IMPORTANT!

This symbol indicates useful information. Not to be used to indicate dangerous situations.

2 DEVICE DESCRIPTION

2.1 Introduction

UNITROL 1000-15 is an automatic voltage regulator of the latest design for synchronous generators and synchronous motors. The unit contains the most advanced microprocessor technology together with IGBT semiconductor technology (Insulated Gate Bipolar Transistor).

A practical and simple-to-operate panel on the unit is used for all control operations. In addition, user-friendly software facilitates commissioning and allows optimisation of operation.

The mechanical construction is extremely compact and robust.

The UNITROL 1000-15 unit can be connected to a 40 A Power Module called UNITROL 1000-PM40. For further information please refer to the separate User's Manual.

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2.2 Area of application

This advanced-design automatic voltage regulator is used for the excitation of indirectly excited synchronous machines. This unit is only suitable for this one area of application.

The regulator can also be switched over to function as a reactive power-, power factor- or field current regulator.

SM = Synchronous Machine

E = Exciter

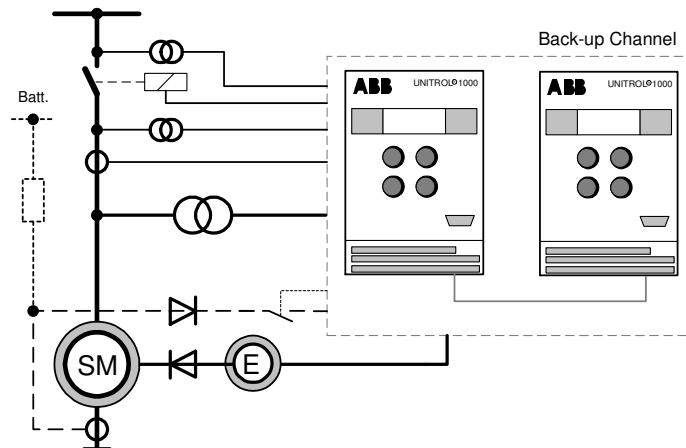
PMG = Permanent-Magnet-Generator

Shunt excitation with:

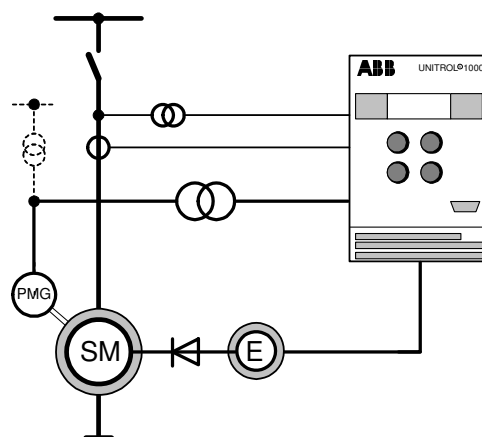
- short circuit support (Boost)

Optional:

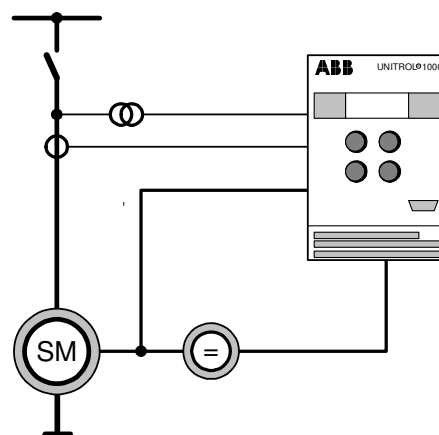
- Power System Stabilizer (PSS)
- Synchronization unit
- Diode monitoring
- Change over to a back-up unit.



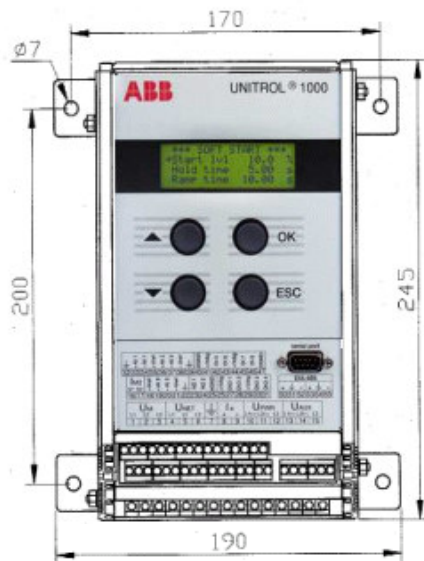
Generator or motor excitation with PMG or external supply.



Replacement of voltage regulators for generators or motors with direct-current excitation machines.



2.3 Hardware



Dimensions in mm

Structure:

The device is built into an aluminium casing with cooling flanges.

The connection terminals are integrated into the front of the circuit boards, which are arranged on top of one another.

Power electronics:

The power part is fitted with an IGBT semiconductor. The average value of the output voltage is always positive. The output is current-limited and thus short-circuit-proof.

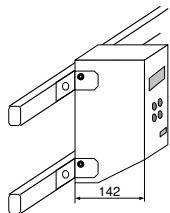
Control elements:

The operating keys, the display and the sub-D connector for the RS-232 interface are located on the unit cover.

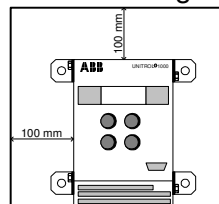
Installation:

The site of installation must be dry and free of dust.

Rack mounting



Wall mounting

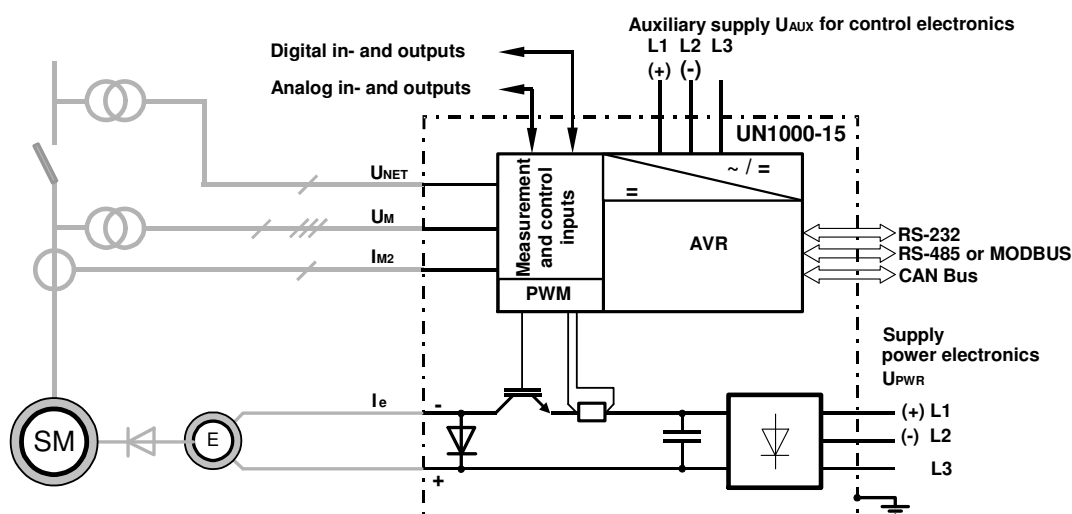


Mounting:

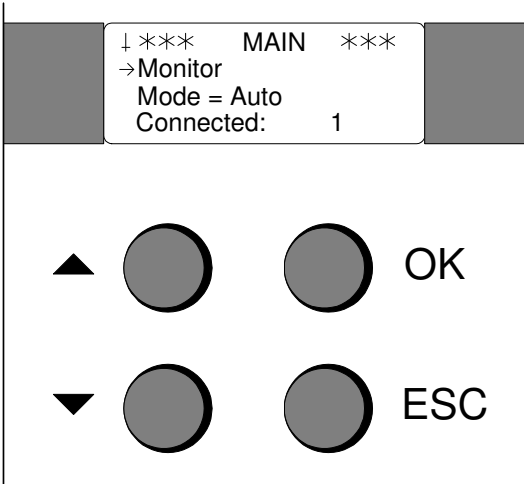
The UNITROL 1000-15 is designed for rack or wall mounting in upright position.

For optimal cooling is to keep free a distance from approx. 100 mm around the unit.

Connection diagram:



2.3.1 Control elements and interfaces



Carrying out settings on the unit

The four-line display and the four keys are sufficient to allow complete operation. All settings can be carried out directly on the unit without additional equipment:

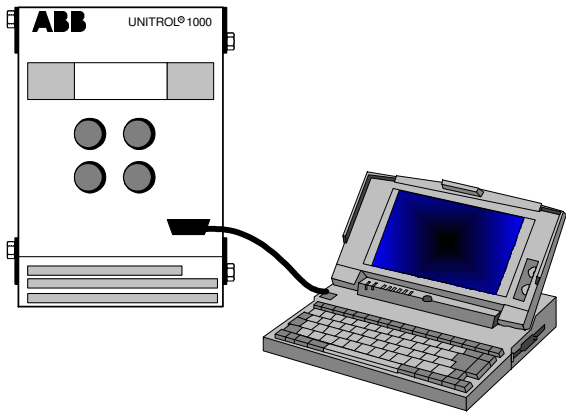
- Configuration of inputs and outputs
- Parameter setting
- Display of important measuring values.

Interface with PC

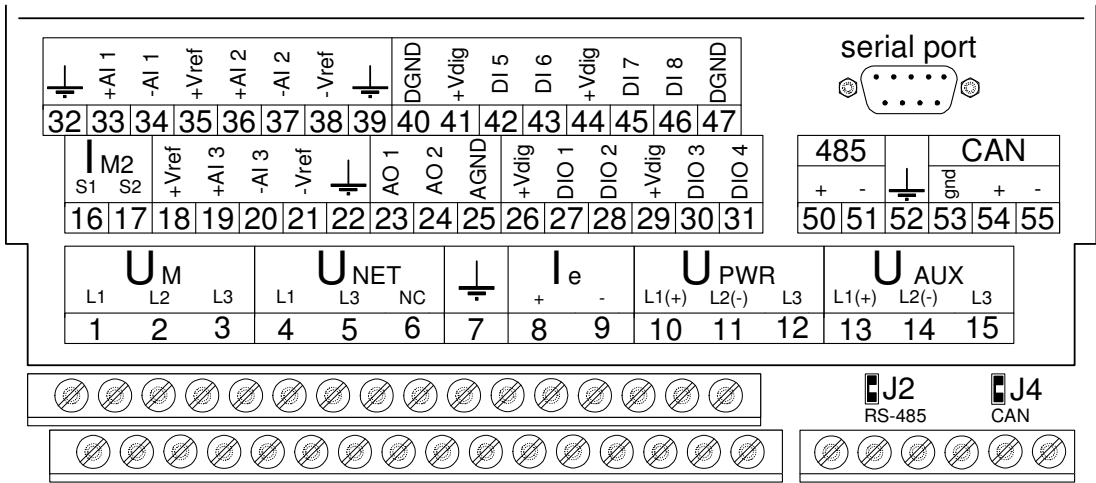
Parameter setting and also optimisation is possible using the user-friendly software *CMT1000* for Microsoft Windows.

Connection cable, sub-D, 9 pol (female)

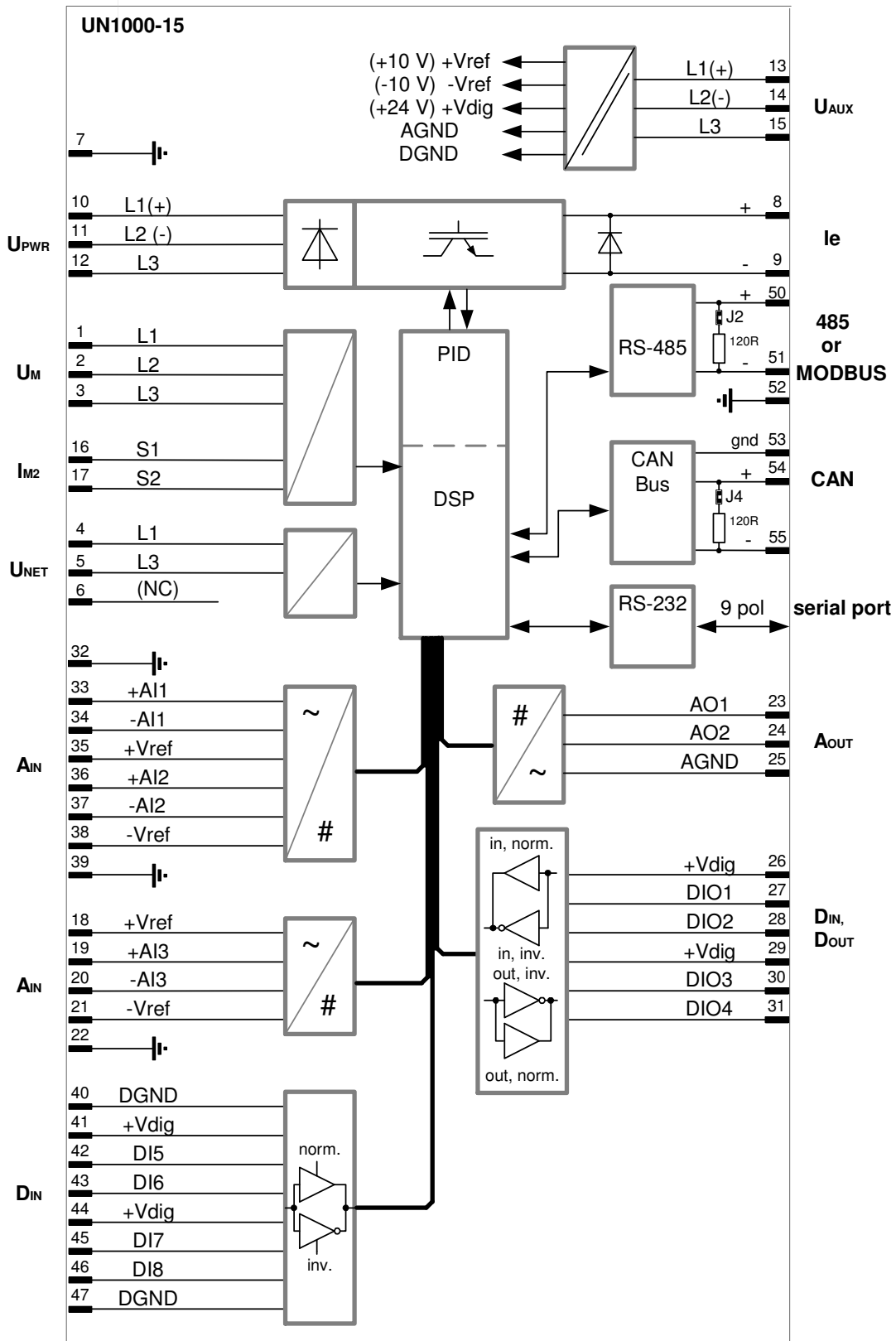
- Configuration of inputs and outputs
- Parameter setting
- Trending function for optimisation of the controller (Oscilloscope, Power chart)
- Display of important measuring values
- Parameter File upload or download.



Terminal block



2.3.2 Overview of the device connections



2.3.3 Device connections

Terminal designation	Signal	Circuit
L1(+) = 13 L2(-) = 14 L3 = 15	<u>Auxiliary supply U_{AUX}</u> - AC input voltage three-phase (250 V / 140 V) or - AC input voltage single-phase (250 V) or - DC input voltage	External 9 to 250 V~ 3~ L1, L2, L3 UN1000-15 50 to 250 V~ 1~ L1, L2 UN1000-15 18 to 300 V= 2= L1(+), L2(-) UN1000-15
L1(+) = 10 L2(-) = 11 L3 = 12	<u>Power electronics supply U_{PWR}</u> - AC input voltage three-phase, AC input voltage single-phase and excitation current < 10 A or - DC input voltage - AC input voltage single-phase and excitation current > 10 A Rectifier: 35 A, 1200 V Capacitor: 1000 µF, 600 V	External 0 to 250 V~ 3~ L1, L2, L3 UN1000-15 0 to 250 V~ 1~ L1, L2 UN1000-15 0 to 300 V= 2= L1(+), L2(-) UN1000-15 External 0 to 250 V~ L1 L2 UN1000-15
Three-phase: L1 = 1, L2 = 2, L3 = 3 S1 = 16, S2 = 17 Single-phase: L1 = 1, L3 = 3 S1 = 16, S2 = 17	<u>Measurements inputs</u> Three-phase: - Machine voltage U _m Single-phase: - Machine current I _{m2} or Single-phase: - Machine voltage U _m or Three-phase with ground (Three ph gnd) - Machine voltage U _m	External L1 L2 L3 U_m max. 250 V / 0.2 VA UN1000-15 SM IM2 S1 S2 1 A / 0.1 VA External L1 L3 U_m max. 150 V / 0.2 VA UN1000-15 External L1 L2 L3 U_m max. 150 V / 0.2 VA UN1000-15
L1 = 4, L3 = 5	Single-phase: Line voltage measurement U _{NET}	External L1 L3 U_{NET} max. 150 V / 0.2 VA UN1000-15
+ = 8 - = 9	Excitation current output I _e	UN 1000-15 External 0 to 300 V= 15 A = E
Earth connection = 7	Earth connection	

Terminal designation	Signal	Circuit
DIO1 = 27 DIO2 = 28 DIO3 = 30 DIO4 = 31 +Vdig = 26, 29	<u>Digital inputs</u> DIO1 to DIO4 DIO1 to DIO4 can be operated either as inputs or outputs. <u>Digital outputs</u> DIO1 to DIO4 Caution: Configured as outputs, DIO1 to DIO4 must not be connected directly with +Vdig (causes short circuit via internal transistor)	
DI5 = 42 DI6 = 43 DI7 = 45 DI8 = 46 +Vdig = 41, 44 DGND = 40, 47	<u>Digital inputs</u> DI5 to DI8 Controlled from potential-free contacts	

The internal 24 V supply may be loaded with a maximum of 300 mA by all used digital inputs and outputs.

Terminal designation	Signal	Circuit
+AI1 = 33, -AI1 = 34 +AI2 = 36, -AI2 = 37 +AI3 = 19, -AI3 = 20 +Vref = 18, 35 -Vref = 21, 38 AGND = 25 Earth = 22,32,39	<u>Analog inputs</u> AI1 to AI3 R = 10 kΩ, 1 W	
+AI1 = 33, -AI1 = 34 +AI2 = 36, -AI2 = 37 +AI3 = 19, -AI3 = 20 +Vref = 18, 35 -Vref = 21, 38 +Vdig = 26, 29, 41, 44 AGND = 25 DGND = 40, 47 Earth = 22,32,39	<u>Analog inputs digitally assigned</u> AI1 to AI3 (DI9 to DI14) see chapter 2.3.6	
AO1 = 23 AO2 = 24 AGND = 25	<u>Analog outputs</u> AO1 to AO2	
Serial port: Sub-D, 9 pol, Male	Serial interface RS-232 RX = 2 TX = 3 GND = 5	Standard cable (terminals 2 and 3 crossed)
485: += 50, -= 51 MODBUS	- Serial interface RS-485 - or MODBUS	
CAN: += 54, -= 55 gnd = 53	CAN Bus for Power Module UNITROL 1000-PM40 Shielded cable: 2x 2x 0.34 mm ² , Impedance 120 Ω, Length of cable < 30 m	
Earth: = 52		

2.3.4 Digital inputs

Input function	Description												
None	Input not assigned												
Excitation ON active	Excitation ON demand active: - Field flashing begins if <i>Off Level</i> > 0% - Soft start begins after the <i>Off Level</i> has been reached (Auto mode)												
Excitation ON not active	Excitation ON demand not active: All setpoints are immediately set to the following values and remain fixed there. <table border="1"> <thead> <tr> <th>Mode</th><th>Preset value</th></tr> </thead> <tbody> <tr> <td>Manual</td><td>0%</td></tr> <tr> <td>Open Loop</td><td>0%</td></tr> <tr> <td>PF</td><td>1.0</td></tr> <tr> <td>Var</td><td>0%</td></tr> <tr> <td>Auto</td><td>100%</td></tr> </tbody> </table>	Mode	Preset value	Manual	0%	Open Loop	0%	PF	1.0	Var	0%	Auto	100%
Mode	Preset value												
Manual	0%												
Open Loop	0%												
PF	1.0												
Var	0%												
Auto	100%												
Gen CB Closed Status active	Circuit-breaker closed status active: - Activates current measurement - This message triggers immediately the ramp of the soft start as if still the hold time												
Gen CB Closed Status no longer active	Circuit-breaker closed status changes from active to not active: All setpoints are immediately set to the following values. <table border="1"> <thead> <tr> <th>Mode</th><th>Final value</th></tr> </thead> <tbody> <tr> <td>Manual</td><td>90% I_e No Load</td></tr> <tr> <td>Open Loop</td><td>90% 1/K_{ceil}</td></tr> <tr> <td>Auto</td><td>100%</td></tr> </tbody> </table>	Mode	Final value	Manual	90% I _e No Load	Open Loop	90% 1/K _{ceil}	Auto	100%				
Mode	Final value												
Manual	90% I _e No Load												
Open Loop	90% 1/K _{ceil}												
Auto	100%												
Parallel with Grid Status	Parallel with grid status active: - Enables together with Gen CB Closed possibility to change over to PF and Var modes. - Disables VDC mode.												
Increase	Increase setpoint of active regulator												
Decrease	Decrease setpoint of active regulator												
Reset Setpoint	Status reset setpoint is active: The setpoint of the active regulator goes to the following value at ramp speed. <table border="1"> <thead> <tr> <th>Mode</th><th>Final value</th></tr> </thead> <tbody> <tr> <td>Manual</td><td>I_e No Load</td></tr> <tr> <td>Open Loop</td><td>100% 1/K_{ceil}</td></tr> <tr> <td>PF</td><td>1.0</td></tr> <tr> <td>Var</td><td>0%</td></tr> <tr> <td>Auto</td><td>100%</td></tr> </tbody> </table>	Mode	Final value	Manual	I _e No Load	Open Loop	100% 1/K _{ceil}	PF	1.0	Var	0%	Auto	100%
Mode	Final value												
Manual	I _e No Load												
Open Loop	100% 1/K _{ceil}												
PF	1.0												
Var	0%												
Auto	100%												
Remote SP Enable	Control signal for remote setpoint setting. The setpoints are controlled by analog inputs.												
PF Enable	Power factor regulation active												
Var Enable	Reactive power regulation active												
Manual Enable	Manual active, field current regulation												
Open Loop Enable	Open loop, direct control of power transistor active												
Synchronize	Synchronize demand active												
VDC Enable	Voltage droop compensation control active												
Secondary Net	Net 2 on island operation (VDC)												

Input function	Description
Reset Alarm	Clear the following alarms: - System OK (not available) - SW Alarm
Standby	Standby demand active: - The output of the AVR is turned off - Boost is disabled - Field flashing is off - Integrator is kept in 1/Kceiling - All limits are disabled - There will be no data transmission to the RS-485 bus - Mode is standby - All setpoints are following the corresponding actual value
RC Fieldbus Block	Disable the Fieldbus communication in local operation
FCB closed Status	Field circuit breaker closed status active

The assignment of the digital inputs to the terminals **DIO1 to 4**, **DI5 to 8**, (**DI9 to 14**) can be selected freely.

Polarity of the digital input

Software setting



2.3.5 Digital outputs

Output function	Description
None	Output not assigned
Boost	Status signal boost is active Boost supports excitation in the event of line short circuit or heavy load. The boost function is blocked during field flashing and soft start.
Field Flashing See section 2.2.2	Field flashing (voltage built up) active, if Excitation ON The next field flashing can only be started after Excitation ON or after the power has been switched off. During field flashing the output of the regulator is blocked in all operation modes.
System OK (optional)	In preparation
Limit Active	One of the limiters (V/Hz, I _e , PQ or UM) is active or setpoint limit has been reached (min. - or max. position)
V/Hz Limit Active	V/Hz limiter active
SP Limit Reached	Setpoint limit has been reached
SP Minimum Reached	Minimum setpoint has been reached
SP Maximum Reached	Maximum setpoint has been reached
Operational Limit Active	I _e , PQ, UM limiter or Diode Alarm active
Min I _e Active	I _e minimum current limiter active
Max I _e Active	I _e maximum current limiter active
Min PQ Active	PQ limiter active
Min UM Active	Limit value for minimum machine voltage is fallen below, voltage limiter active
Max UM Active	Limit value for maximum machine voltage is exceeded, voltage limiter active
Voltage Relay	Active = machine voltage below boost threshold Inactive = boost threshold and hysteresis exceeded Not dependent on signal <i>Excitation ON</i> .
Close CB Command (Option)	The command is released: - Angle is in a value where it would take the <i>Total CB Closing Time</i> for the breaker to close at zero degrees with current speed and acceleration. - Sync Check demand active - The command is so long active as the command Sync Check is active
Sync Check (Option)	Sync Check demand active: - Generator circuit breaker is open - Machine voltage is more than 50% - Synchronization is possible (unit with Sync-Option) - Synchronize is enabled - Slip is between minimum slip and maximum slip - Machine voltage is deviating from network voltage maximum Delta U - Angle is between -maximum delta angle and +maximum delta angle

Output function	Description
SW Alarm	Software program process is out of order
Diode Alarm (Option)	Event of open diode
Diode Trip (Option)	Event of a shorted diode
SW Alarm or Diode Trip (Option)	SW alarm and event of a shorted diode
MODBUS Com Alarm	Loss of MODBUS command
Close FCB Command	Command to close the field circuit breaker
Open FCB Command	Command to open the field circuit breaker

The assignment of the digital outputs to the 4 terminals **DIO1 to 4** can be selected freely.

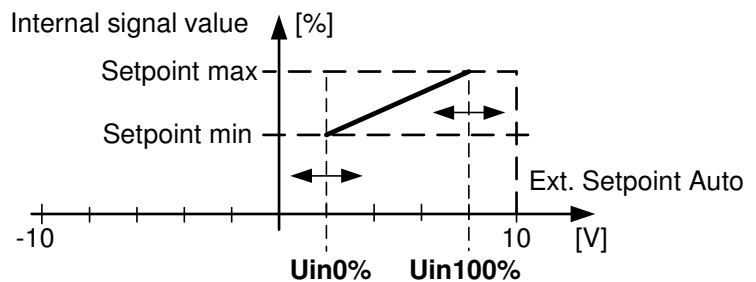
2.3.6 Analog inputs

Input function	Description
None	Input not assigned
Auto Remote Setpoint	External setpoint input to Auto regulator
PF Remote Setpoint	External setpoint input to PF regulator
Var Remote Setpoint	External setpoint input to Var regulator
Manual Remote Setpoint	External setpoint input to Manual regulator
Open Loop Remote SP	External setpoint input to open loop
UM Aux	Auxiliary supply to the summing point of Auto regulator
Reserved2	Reserved for later function expansion
Cooling Media Temperature	Input temperature measurement for the temperature influence limiter
Reserved4 to 5	Reserved for later function expansion
Digital Input 9(+) & 10(-)	Assign digital inputs
Digital Input 11(+) & 12(-)	Assign digital inputs
Digital Input 13(+) & 14(-)	Assign digital inputs

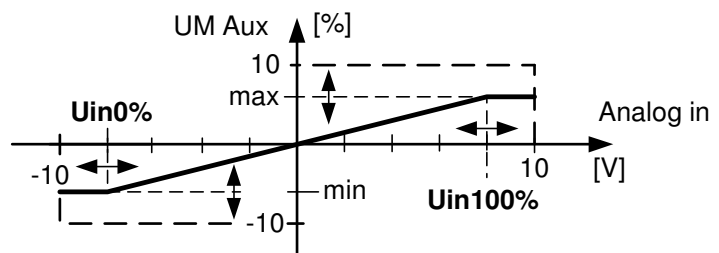
The assignment of the analog inputs to the 3 terminals **AI1 to 3** can be selected freely.

Level of the analog inputs, see parameter settings chapter 7.3

- External setpoint input



- Input to the summing point



- Input voltage of the cooling media temperature
 - Range: Uin0% = -10.0 V refers to -100 °C
 - Uin100% = 10.0 V refers to 100 °C

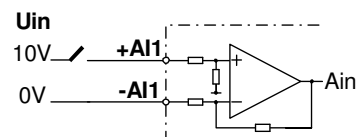
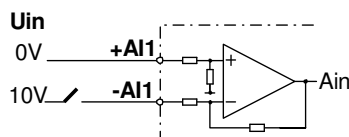
- Analog inputs digitally assigned
Two digital inputs can be generated from one analog input. See this example.

Voltage between the input terminals

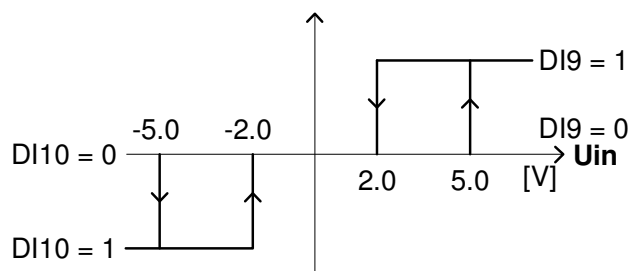
$$(+)\text{AI1} - (-)\text{AI1} \leq -5 \text{ V}$$

$$(+)\text{AI1} - (-)\text{AI1} \geq 5 \text{ V}$$

Principle hardware configuration



Condition of the digital inputs



Status of the digital inputs

-> Din9(+) = 0
-> Din10(-) = 1

-> Din9(+) = 1
-> Din10(-) = 0

+AI or -AI shall not be active simultaneously.

Configuration of the inputs signals with the CMT-Tool:

Select the analog input AI1 and adjust the input range
[Setup \ Analog Inputs]

Select the two digital signals
DI9 from +AI
DI10 from -AI
[Setup \ Digital I/Os]

Direction = In

- None
- Auto Remote Setpoint
- PF Remote Setpoint
- Var Remote Setpoint
- Manual Remote Setpoint
- Open Loop Remote Setpoint
- UM Aux
- Reserved2
- Cooling Media Temperature
- Reserved4
- Reserved5
- ☒ Digital Input 9(+) & 10(-)
- Digital Input 11(+) & 12(-)
- Digital Input 13(+) & 14(-)

- Range: $U_{in0\%} = 2.0 \text{ V}$
 $U_{in100\%} = 5.0 \text{ V}$

Direction = In

- ☒ None
- Excitation ON
- Gen CB Closed Status
- Parallel with Grid Status
- Increase
- Decrease
- Reset Setpoint
- Remote SP Enable
- PF Enable
- Var Enable
- Manual Enable
- Open Loop Enable
- Synchronize
- VDC Enable
- Secondary Net
- Reset Alarm
- Standby
- RC Fieldbus Block
- FCB closed Status

2.3.7 Analog outputs

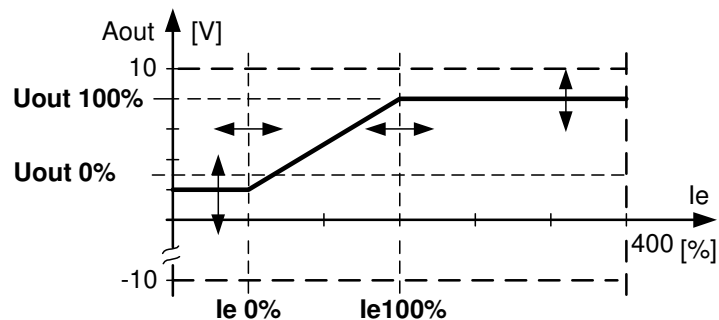
Output function	Description
None	Output not assigned
Excitation Current	Excitation current
PWM	Pulse width modulation, control value is limited to minimum of 2%
Fbias	Analog continuous signal presenting difference between UNET and UM $F_{bias} = f_{NET} - f_{NOM} - (Slip_{max} - Slip_{min})/2$ Output signal Fbias is forced to zero: - Not in Sync mode - Synchronization function not enabled (unit without Sync-Option) - Network frequency is not between 45 to 66 Hz

The assignment of the analog outputs to the 2 terminals **AO1 to 2** can be selected freely.

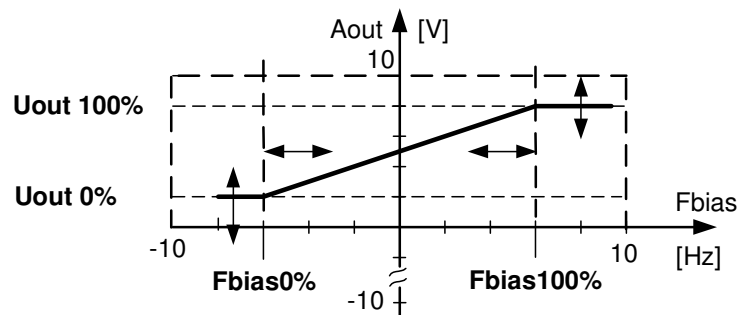
Level of the analog outputs, see parameter settings chapter 7.3

- Excitation Current

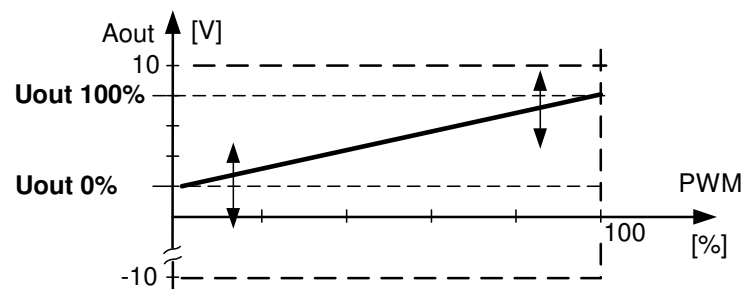
Caution: $I_{e0\%}$ must be less than $I_{e100\%}$



- Fbias



- PWM Output



The full voltage range runs from -10 V to 10 V at a resolution of 10 bits.

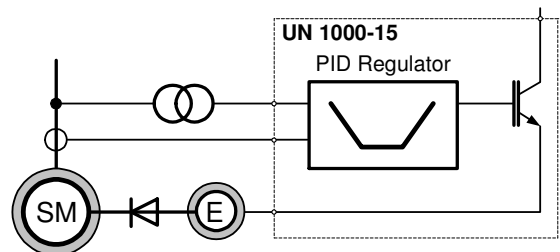
2.4 Software

2.4.1 Operating modes

Bump less changeover between all modes of operation

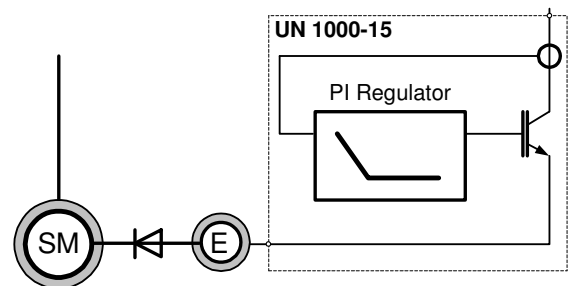
Automatic voltage regulation (Auto)
Regulates the terminal voltage of the synchronous machine.

Note: Current measurement for compensation / droop

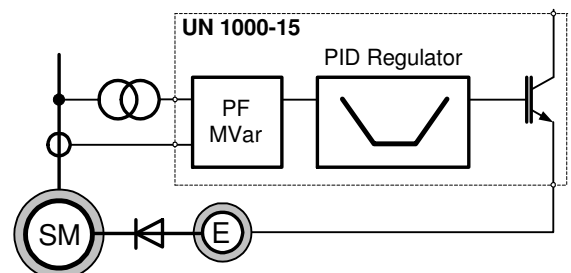


Manual control
Regulates the field current of the excitation machine.

Note: No limiters will be active.

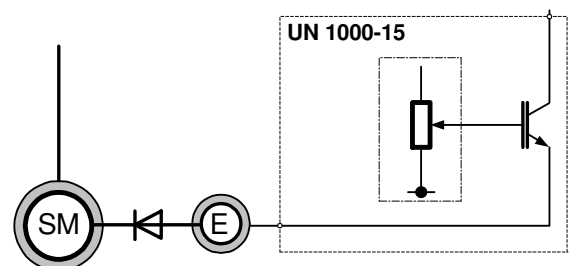


PF or Var regulation (PF, Var)
Regulates the power factor or reactive power of the synchronous machine.



Open loop
Manual control with a fixed output signal.

Note: No limiters will be active.



Setting of operating modes see chapter 5.2, Setting aids

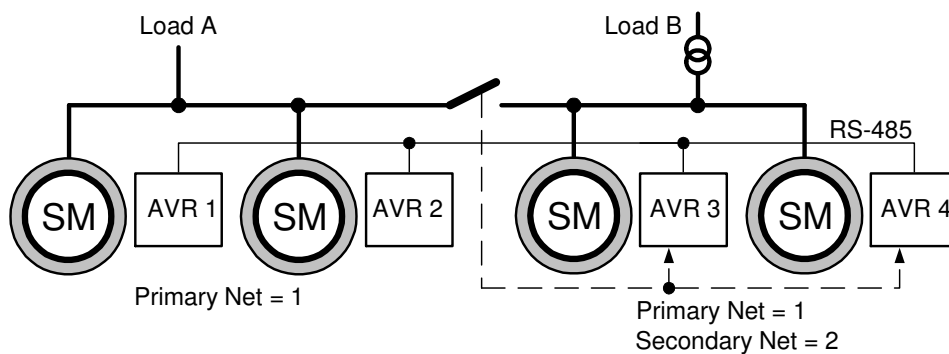
Voltage droop compensation control and reactive power allocation

(VDC)

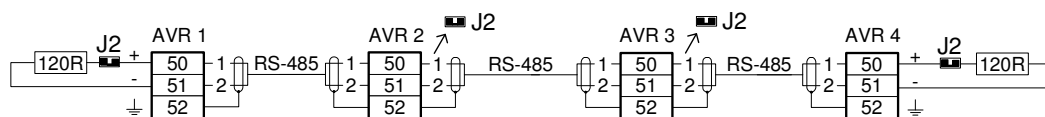
- For island operation only -

In order to share equally the amount of reactive power between parallel connected generators UNITROL 1000-15 offers a special feature, called voltage droop compensation control. Units are connected together with RS-485 bus.

In VDC mode, actually, all AVRs operate in Auto mode with voltage droop. However, each AVR (with its own ID number) writes the value of its own amount of reactive power to the RS-485 bus. Each unit reads these values and calculates a common average Mvar setpoint, and compensates the effect of voltage droop. Therefore, voltage level on the busbar is always kept at 100% (not adjustable).



Connections RS-485:



Cable cross-section = $2 \times 0.75\text{mm}^2$

Cable impedance = 100 to 120R

Shielded twisted pair

When operating in VDC mode, the network can be divided to smaller entities. Each generator can operate in two pre-determined island grids, called Primary Net and Secondary Net.

Secondary net is selected by means of a digital input. If this input is not active, primary net is selected. Each net (island grid) shall have its own network number.

Synchronization

(Sync)

The automatic synchronization of a synchronous machine with the line is achieved using the optional integrated synchronizing function. By setting few simple parameters, UNITROL 1000-15 supplies the corresponding control signals for the speed governor and closes the circuit breaker.

Analogue speed correction Fbias is given out from UNITROL 1000-15 analogue output, to speed governor control summing point (not as pulse). This signal represents the difference of **network** NOMINAL and **network** ACTUAL frequency. The reference (setpoint) value for the speed governor must be nominal (50 or 60 Hz) and the Fbias given by UNITROL 1000-15 will drive the speed close to actual network frequency.

Note that the speed control is not with INC/DEC pulses, and that is not possible to derive such pulses from the Fbias signal. Also, the configuration possibilities of PT's (machine/net) are very limited. See chapter 2.4.2. Other functions, like dead bus synchronising, etc. are also missing. The circuit breaker closing command is traditional type with no specialities (for one breaker).



CAUTION!

Power circuit breaker (CB) may only be closed if both voltages are at least approximately synchronous (coincident).

Otherwise, faults in line operation, loading of the synchronous machine and, in extreme cases, damage to the synchronous machine can result.

A separate synchrocheck relay must be used to secure the right operation. For further information please contact ABB.

Measuring:

The values are formed from the two measurement signals U_{NET} and U_M

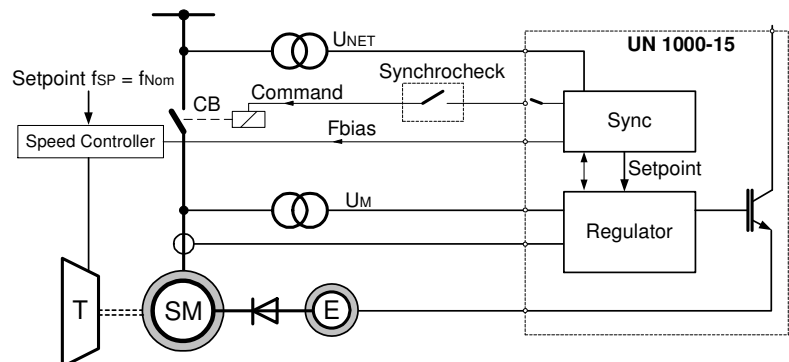
- Voltage difference (amplitude)
- Slip (frequency difference)
- Phase-angle difference

Matching:

The voltage matching function gives the adjusting value to the internal voltage regulator and the frequency matching function send the analog signal Fbias to the turbine regulator.

Monitoring and command generation:

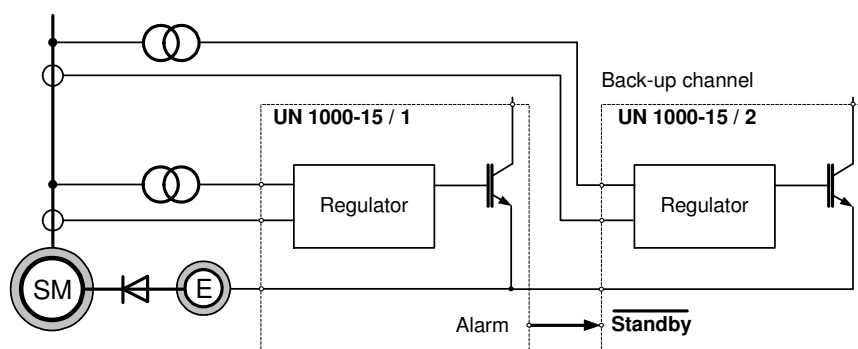
The command to close the breaker (CB) is released if all conditions are fulfilled.



- After the power circuit breaker has been closed the Synchronize command must be deactivated.
- The setpoint of the speed controller must have the nominal value of 50 resp. 60 Hz for the synchronization.

Back-up channel system

In case of a failure in channel 1, the back-up channel 2 is activated and the defective channel switched off. Change over to channel 1 can be controlled by alarm signal or with an external control signal.



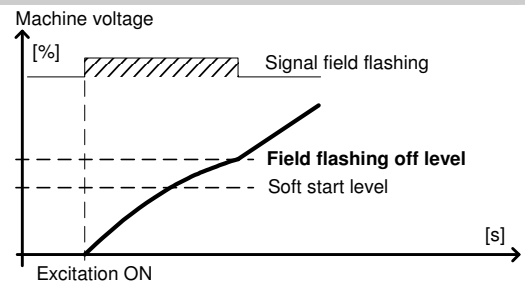
Activating the predefined digital inputs chooses operating modes. See section 5.2.

For further information how to design a redundant system please contact ABB.

Field Flashing

- Off Level [%]

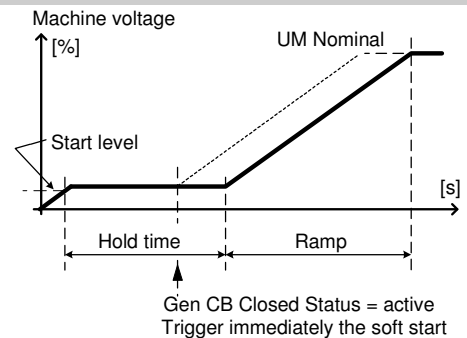
By setting Off level to 0% the field flashing is disabled



Soft Start	
-------------------	--

- | | |
|----------------|-----|
| Starting Level | [%] |
| Hold Time | [s] |
| Ramp Time | [s] |

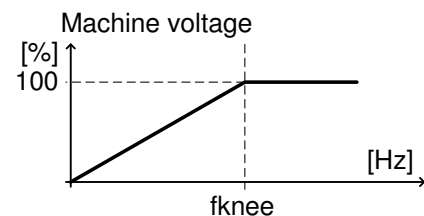
A soft start is done only in the Auto mode



Limiters	
----------	--

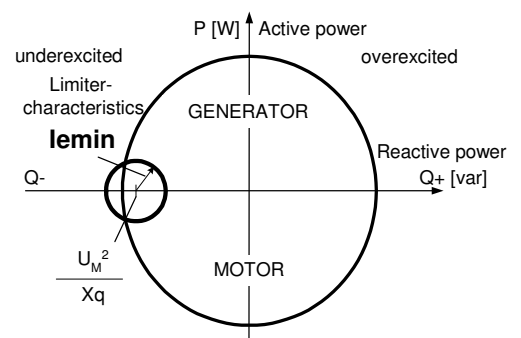
V/Hz Limiter Machine voltage

- | | |
|-------|------|
| fknee | [Hz] |
| Slope | [%] |



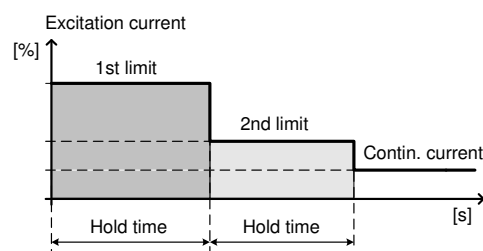
Minimum current limiter

- Minimum [%]
Active = True/False



leMaximum current limiter

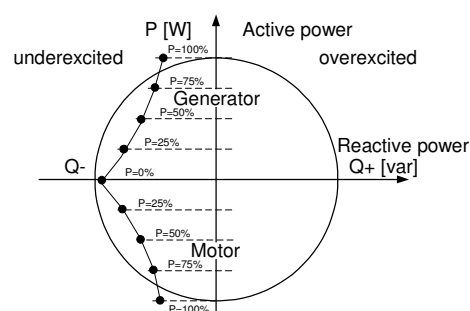
- 1 st limit	Maximum [%]
- Hold time	Maximum Hold Time [s]
- 2 nd limit delay	Delayed [%]
- Hold time	Delayed Hold Time [s]
- Continuous current	Continuous [%]
- Limiter active	Active = True/False



PQ Limiter

The characteristic is determined by 5 points

- Q limit at P= 0%	Minimum Q(P @ 0%) [%]
- Q limit at P= 25%	Minimum Q(P @ 25%) [%]
- Q limit at P= 50%	Minimum Q(P @ 50%) [%]
- Q limit at P= 75%	Minimum Q(P @ 75%) [%]
- Q limit at P= 100%	Minimum Q(P @ 100%) [%]
- Voltage dependency active	Volt.Dependency=True/False
- Limiter active	Active = True/False



UM Limiter (only in PF / Var mode)

- Limiting value of the minimum machine voltage
- Limiting value of the maximum machine voltage
- Limiter active
- Limiter active

Minimum [%]
Maximum [%]
Minimum Active = True/False
Maximum Active = True/False

IM Limiter

- Limiting value of the maximum machine current
- Time Multiplier according to IEC 60255-3, Table 1, Col. B, very inversed characteristic
- Limiter active

Maximum [%]
Time Multiplier K

Maximum Active = True/False

Boost-Output

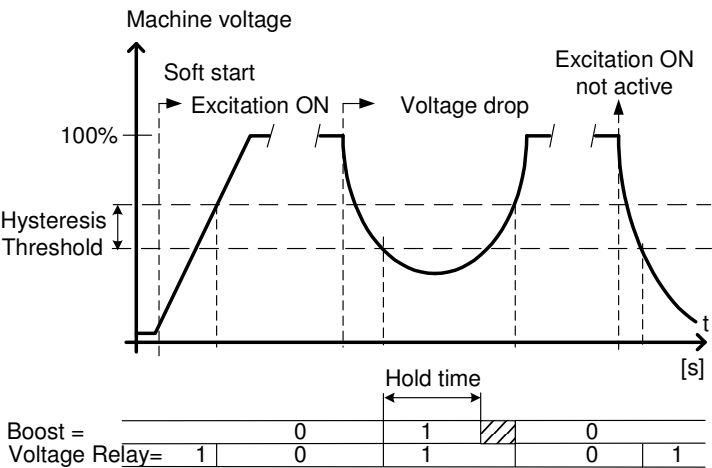
- Pick-up volt. to generate boost command

- Time boost output is active

- Hysteresis for the reset of boost output
- Threshold [%]

Hold Time [s]

Hysteresis [%]



Limiters Temperature Influence

- IM Limiter

T1 at Max IM

T2 at IM th2

T3 at IM th3

T4 at IM = 0 A

Machine current level

Machine current level
- T1

T2

T3

T4

IM th2

IM th3
- [°]

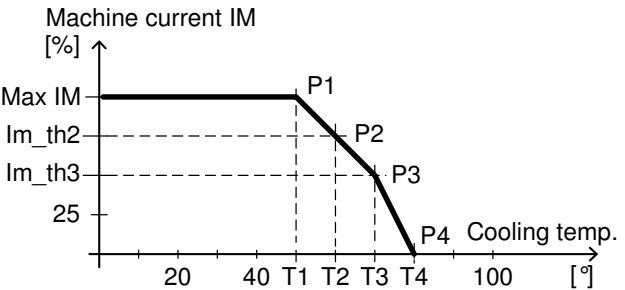
[°]

[°]

[°]

[%]

[%]



- Ie Limiter

T1 at Max Continuous Ie

T2 at Ie th2

T3 at Ie th3

T4 at Ie = 0 A

Excitation current level

Excitation current level
- T1

T2

T3

T4

Ie th2

Ie th3
- [°]

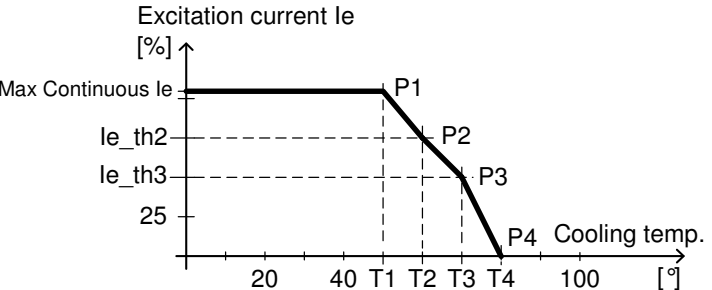
[°]

[°]

[°]

[%]

[%]



Setpoint setting for the regulator

- Auto to voltage regulator
- PF, Var to power factor, reactive power regulator
- Manual, to manual regulator
- Open Loop to open loop regulation circuit

All setpoints have the following parameters

- Minimum
- Maximum
- Ramp Rate

Level of the setpoint if the digital input *Excitation ON* is not active.

Mode	Preset value
Manual	0%
Open Loop	0%
PF	1.0
Var	0%
Auto	100%

The limits and ramp rate can be set separately for each operating mode.

The setpoints of the non-active regulators follow the relevant operating point. For example, for reactive power regulation (Var) the setpoint of the auto-regulator follows the current machine voltage. Thus allows surge-free switching between operating modes if the new setpoint is within the setpoint limit.

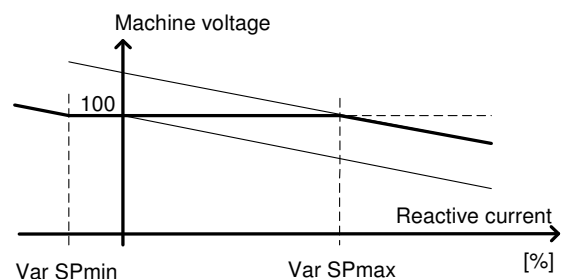
Particulars to the setpoint in Sync mode

- No Load: Regulator works in AUTO mode. Setpoint follows line voltage and the ramp rate corresponds with the increase- resp. decrease command.

Voltage Droop Compensation Control (VDC)

- Primary Net ID Primary Net ID
- Secondary Net ID Secondary Net ID
- Ramp Time Ramp Up Time [s]

Ramp time to get machine voltage to 100% and Var equal between machines after VDC is activated.



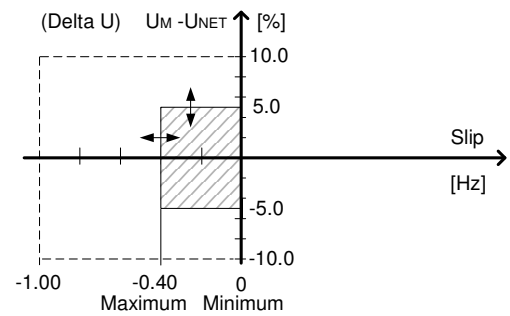
Synchronization

- Minimum Slip
- Maximum Slip
- Max. voltage difference

Min Slip
Max Slip
MaxDeltaU

[Hz]
[Hz]
[%]

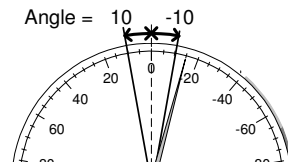
Note: $f_M > f_{NET}$
Slip always negative



Delta Angle

Max Delta Angle

[deg]



- Time between closing command and closed breaker (see data sheet of the breaker)

Tot CB Close Time

[ms]

- Correction factor voltage
- Correction factor of the phase shift between U_{PWR} and U_{NET} .

Voltage Offset

[%]

Angle Offset

[deg]

• Voltage matcher

The machine voltage is raised to the level of the line voltage.

Ramp Rate has to be adjusted in the menu Setpoint AUTO

• Frequency matcher

Example:

$f_{NOM} = 50 \text{ Hz}$ (45 Hz < $f_{NET} \leq 54 \text{ Hz}$, **50 Hz**)
(54 Hz < $f_{NET} < 66 \text{ Hz}$, **60 Hz**)

$f_{NET} = 49 \text{ Hz}$

$f_M = 50 \text{ Hz}$

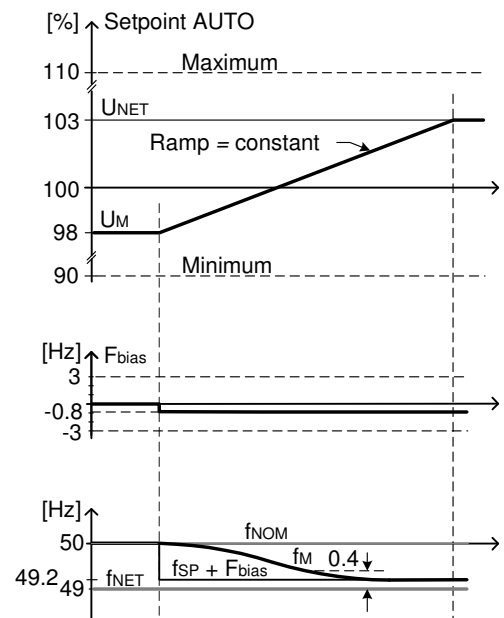
$f_{SP} = 50 \text{ Hz}$ (Setpoint speed governor)

$Slip_{max} = -0.4 \text{ Hz}$

$Slip_{min} = 0 \text{ Hz}$

$F_{bias} = f_{NET} - f_{NOM} - (Slip_{max} - Slip_{min})/2 = -0.8 \text{ Hz}$

$f_{SP} + F_{bias} = 49.2 \text{ Hz}$

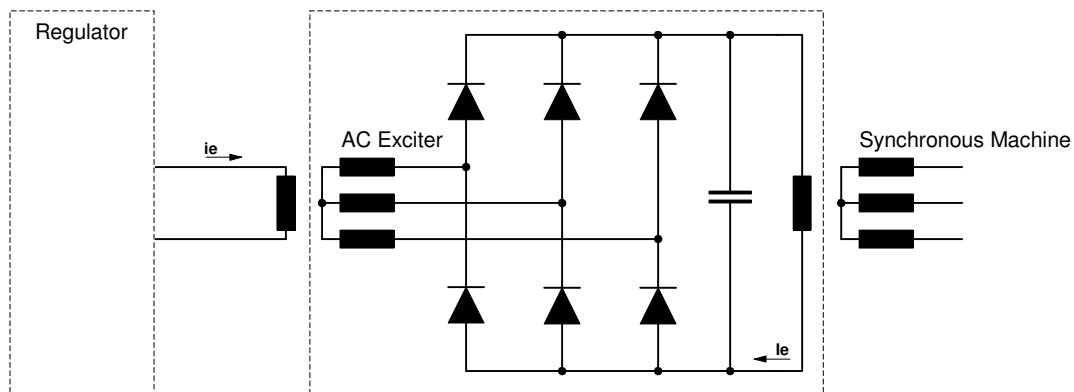


Diode Monitoring

The aim of the Diode Monitoring is detecting the following failures:

- Break of a diode
- Short circuit of a diode

It is needed in brushless excitation systems and can only be done indirectly, because the diodes are part of the rotor. See the principle circuit for the excitation is as follows:



- Nominal exciter frequency (Machine)	f Exc Nominal	[Hz]
- Exciter time constant	Tconst Exc	[s]
- Diode monitoring active	Active =	TRUE/FALSE
- Diode alarm level	Alarm Level	[%]
- Diode alarm delay	Alarm Delay	[s]
- Diode trip level	Trip Level	[%]
- Diode trip delay	Trip Delay	[s]

The device evaluates the alternating current induced in the field circuit of the exciter in the event of a fault in the rotating rectifier.

The ALARM is triggered in the event of a broken branch in the rotating exciter.
The TRIP is triggered in the event of a branch short-circuit in the rotating exciter.

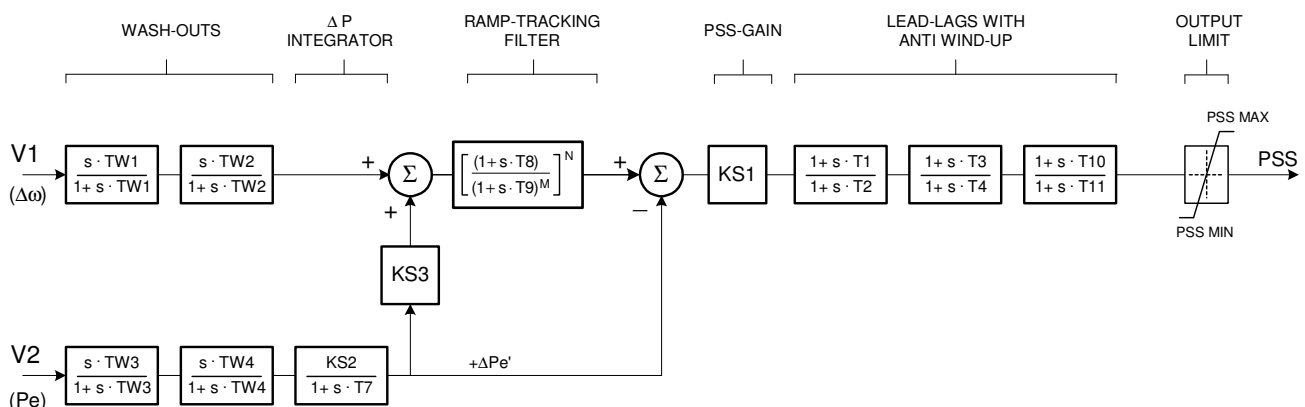
The PSS is used to improve the damping of possible oscillations in large transmission networks by adding a signal to the voltage regulator of the existing excitation system.

It is also used to attenuate local rotor oscillation in synchronous generators through additional influencing of the excitation. The main application is excitation systems in single or multi-machine power plants.

Electromechanical oscillations may be classified in four main categories

- **Local oscillations** between a unit and the rest of the generating station and between the latter and the rest of the power system. Their frequencies typically range from 0.8 to 2.0 Hz.
- **Inter-plant oscillations** between two electrically close generation plants. Frequencies may vary from 1 to 2 Hz.
- **Inter-area oscillations** between two major groups of generation plants. Frequencies are in a typical range of 0.2 to 0.8 Hz.
- **Global oscillation** characterized by a common in phase oscillation of all generators as found on an isolated system. The frequency of such a global mode is typically under 0.2 Hz.

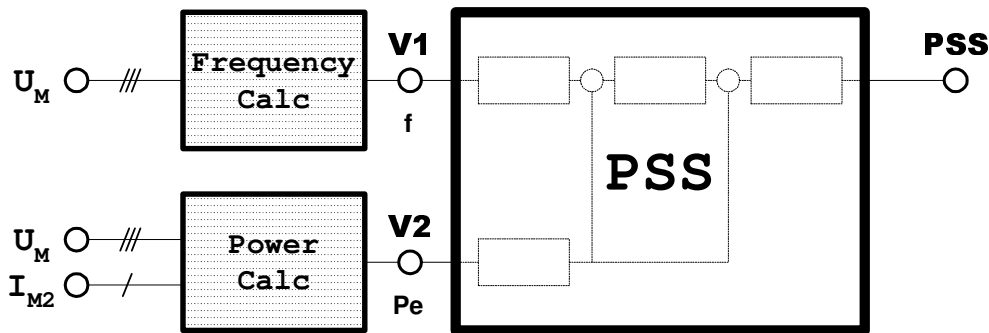
The IEEE Std. 421.5-1995 PSS 2A/2B type model functionally represents the PSS. This model is showed in the following figure.



The object of power system stabilizing (PSS) equipment is to increase the generator exciter's contribution to improving the stability to the highest possible operating range of the generator. The PSS derives additional signals from the generator internal frequency, which considerably improves the stability of the power transmission.

- Dual input (frequency, electrical power).

The following figure shows an simplified diagram of the PSS and related functions.



Due to the PSS function implementation and structure of UNITROL 1000 AVR, there are some points to be noticed when using this PSS. These are elements limiting the maximum performance of the PSS:

- Brushless excitation system,
- Single phase machine current measurement (only) is available,
- Positive field voltage (only) can be supplied.

Parameters

The PSS function can be activated by PSS_SELECT parameter

The parameters will not be explained here. They can be calculated by ABB Switzerland. For further information please see the Functional Description of UNITROL 1000, Power System Stabilizer, 3BHS213239 E01. See manufacturer's details on page 2.

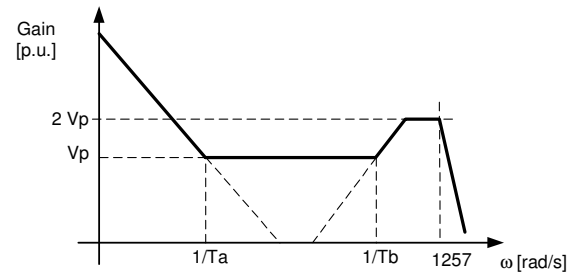
Regulator

Tune

Auto [Voltage control]

- Proportional gain
- Derivative time constant
- Integral time constant

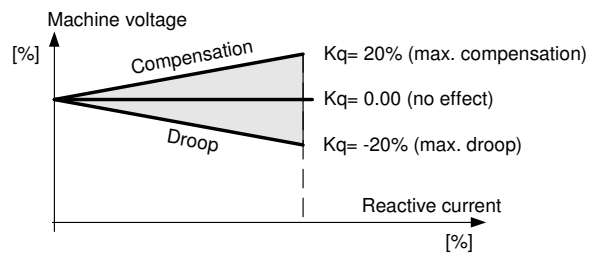
Proportional Gain V_p
Derivation Time T_b [s]
Integration Time T_a [s]



- Compensation or droop

Droop K_q

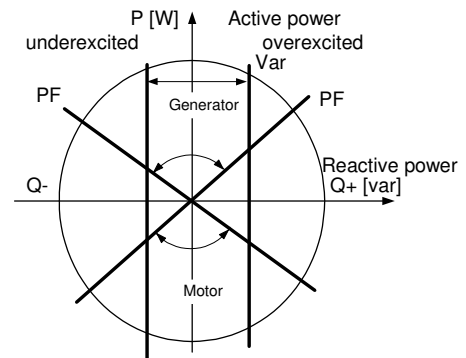
[%]



PF or Var Control and PQ Limiter

Proportional Gain V_p
Derivation Time T_b [s]*
Integration Time T_a [s]

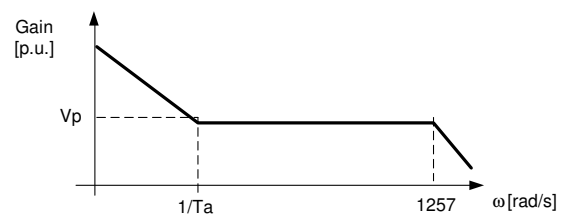
*) Only one parameter, see voltage control



Manual

Field current control and
le Limiter

Proportional Gain V_p
Integration Time T_a [s]



Communication

- ID Definition ID unit number: AVR ID
- Port Configuration
- MODBUS For more information, refer to Modbus Manual

Write EEPROM

Saves the parameters in the device's EEPROM

Electrically Erasable
Programmable Read Only Memory

3 INSTALLATION AND DISPOSAL

The unit should be unpacked with the usual degree of care, without the use of force and using suitable tools.

The unit should be inspected visually to check for any damage caused during transport. Complaints regarding defects resulting from inappropriate transport are to be addressed immediately to the receiving station or the last carrier.



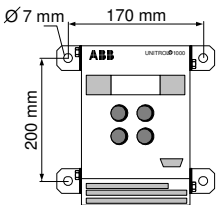
NOTICE!

- The unit is visible damaged.
- The safe operation is not possible.
- The unit must not be installed and taken in operation

3.1 Mechanical installation

The unit is fixed in place by means of four screws.
See dimensions diagram for fixing holes and spacing.
Mounting instructions see page 9.

The unit should only be installed in indoor areas which are dry and dust-free and which do not contain any gases, acid fumes or similar.



3.2 Earthing and wiring

The emission limits in accordance with standard EN 61000-6-4 will only be complied with if the connections for the power electronics supply and the field output are made using shielded cables earthed at each end. We also recommend that shielded cables be used for the analog and digital connections.

The connection terminals are fitted on the front side of the unit, on three levels:

- Cross sections:
- Power electronics, terminal numbers 1 to 15: 0.2 to 4 mm² (AWG 10 to 30)
 - Electronic sections, terminal numbers 16 to 55: 0.2 to 2.5 mm² (AWG 12 to 30)

The casing must be earthed at terminal 7 with 4 mm². Additional earth terminals are provided for the shielded cables. The earth connection should be kept as short as possible.
The tightening torque for all terminals is: 0.6 Nm (5 - 7 lbf. in.).

3.3 Disposal



Improper disposal of electrical devices can lead to environmental damage. It is therefore important that qualified personnel carry out the disposal of electronic equipment.

The metallic casing does not present any risk to the environment and can be recycled. The printed circuit boards are simple to remove. The boards must be removed and should be disposed by a licensed disposal company. Environmentally hazardous elements such as capacitors must be separated from the boards.

ABB	ABB Switzerland Ltd	Document number	Lang.	Rev. ind.	Page
		3BHS201795 E80	en	C	37

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4 OPERATION

4.1 Operation on the unit

4.1.1 Keypad

The keypad consists of four keys with the following functions:

Keys	Commands
▲, ▼	Select mode: Moves the cursor up or down Horizontal arrow cursor marks the active line Parameter setting: Increases or reduces value
OK	Select mode: Selects sub-menu in the line marked with the cursor On parameter level, the cursor disappears and the following text is displayed: OK to write value => Accept new value, not yet saved in EEPROM ESC to cancel => Retain old value Parameter setting: Accept new value
ESC	Select mode: Back to next-higher menu Parameter setting: Retain saved value
ESC+▼ ESC+▲	Keys locked, panel light off Keys unlocked, panel illuminated

4.1.2 Display

After auxiliary voltage supply UAUX is applied the software revision number of the unit (Cntrl) and the panel are shown on the display for a few seconds. Then the start menu *MAIN* lights up.



All the sub-menus are called up from the "MAIN" start menu

The vertical arrow (↓ ↑ ↓) indicates whether other lines are available using the up (▲) or down (▼) keys.

The horizontal arrow cursor (→) marks the active line.

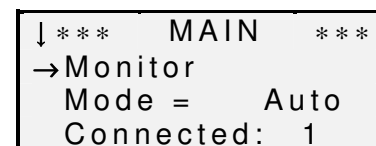


"MAIN" start menu

The title and three lines are displayed

- Online measurement of machine and field values
- Current operating mode, in this place only a information
- Set ID number of the unit, in this place only a information
- Set parameters
- Set regulator parameters
- Set ID number
- Save parameters in UNITROL 1000-15

- ▼ key
- ▼ key
- ▼ key
- ▼ key



Setup
Tune
Communication
Write EEPROM

4.1.3 Example of parameter setting

Example: Assigning *Parallel wGrid* to input DI6

Operating	Key	Resulting display
Display after auxiliary voltage is applied		↓*** MAIN *** →Monitor Mode = Auto Connected: 1
In the start menu, use ▼ key to move down to setup	3x ▼	↓*** MAIN *** Mode = Auto Connected: 1 →Setup
Select sub-menu 1, setup, and use ▼ key to move down to digital I/Os	OK, 7x ▼	↓*** SETUP ** Setpoints Volt.Droop Comp. →Digital I/Os
Select sub-menu 2, digital I/Os, and use ▼ key to move down to I/Os	OK, 1x ▼	DIGITAL I/Os Hysteresis →I/Os Din from Ain
Select sub-menu 3, digital I/Os, and use ▼ key to move down to DI6	OK, 5x ▼	↓ DIGITAL I/Os DIO4 DI5 →DI6
Select sub-menu 4, DI6	OK	*** DI6 *** →Din= None Polarity =Normal
Select parameter DI Cursor disappears, text OK/ESC appears instead	OK	*** DI6 *** Din= None - OK to write value - ESC to cancel
Use ▲ key to select assignment	3x ▲	*** DI6 *** Din=Parallel wGrid - OK to write value - ESC to cancel
Press OK key to save	OK	*** DI6 *** →Din=Parallel wGrid Polarity =Normal
Press ESC key to return to start menu	4x ESC	↓*** MAIN *** Mode = Auto Connected: 1 →Setup

- The above procedure only changes the parameters in the volatile memory. To make the modification permanent, you have to save the entire parameter set:
- Save parameters: Start menu **MAIN**, Write **EEPROM**, **OK**, **OK**.

4.1.4 Menu structure of display

Start menu Sub-menu 1 Sub-menu 2 Sub-menu 3

```
***   MAIN   ***
Monitor
Mode      = Auto
Connected: 1
Setup
Tune
Communication
Write EEPROM
```

```
*** MONITOR ***
UM      = kV
UM rel  = %
IM2     = A
IM2rel  = %
Q       = kVar
Q rel   = %
P       = kW
P rel   = %
Ie      = A
Ie rel  = %
freq    = Hz
```

MAIN ➡

```
***   SETUP   ***
System Data
Field Flashing
Soft Start
Limiters
Synchronization
Setpoints
Volt. Droop Comp.
Digital I/Os
Analog Inputs
Analog Outputs
Diode Monitoring
PSS
```

```
* SYSTEM DATA *
UMNom      = kV
PT         = □
UMPrim     = kV
UMSecond   = V
IM2Nom     = A
IM2Prim    = A
IM2Sec     = A
Ie Nom     = A
Ie NoLoad  = %
Kceil      = V/V
f Nominal  = Hz
Xq         =
```

*PT:Three phase
Single phase
3 phase gnd*

MAIN, SETUP ➡

```
*FIELD FLASHING
Off Level  = %
```

MAIN, SETUP ➡

```
**SOFT START**
Start Lvl  = %
Hold Time  = s
Ramp Time  = s
```

MAIN, SETUP ➡

```
*** LIMITERS ***
V/Hz Limiter
IeMin Limiter
IeMax Limiter
PQ Limiter
UM Limiter
IM Limiter
Boost
Temp Influence
```

** V/Hz LIMITER *
fknee = Hz
Slope = %*

Sub-menu 2

Sub-menu 3

MAIN, SETUP,

LIMITERS 

* leMin LIMITER *
Minimum = %
Active =

MAIN, SETUP,

LIMITERS 

* leMax LIMITER *
Maximum = %
Hold Time = s

Delayed = %
Hold Time = s

Continuous = %
Active =

MAIN, SETUP,

LIMITERS 

** PQ LIMITER **
Q(0%) = %
Q(25%) = %
Q(50%) = %
Q(75%) = %
Q(100%) = %
Volt.Depend =
Active =

MAIN, SETUP,

LIMITERS 

** UM LIMITER **
Minimum = %
Maximum = %
MinActive =
MaxActive =

MAIN, SETUP,

LIMITERS 

** IM LIMITER **
Maximum = %
Time Mult =
Active =

MAIN, SETUP,

LIMITERS 

*** BOOST ***
Threshold = %
Hold Time = s
Hysteresis = %

MAIN, SETUP,

LIMITERS 

Sub-menu 4

*IM Limit Derating
T1 =
T2 =
IM th2 = %
T3 =
IM th3 = %
T4 =

Temp Influence
IM Limiter Derating
le Limiter Derating



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Sub-menu 2

Sub-menu 3

Sub-menu 4



*le Limit Derating		
T1	=	
T2	=	
le th2	=	%
T3	=	
le th3	=	%
T4	=	

MAIN, SETUP ➡

SYNCHRONIZATION		
Min Slip	=	Hz
Max Slip	=	Hz
Max dU	=	%
Max Angl	=	deg
TotCBCtime	=	ms
U Offset	=	%
Angle Off	=	deg

MAIN, SETUP ➡

** SETPOINTS **		
Auto		
PF		
Var		
Manual		
Open Loop		

** AUTO		
** PF SP		
** Var SP		
** MANUAL SP		
* OPEN LOOP SP *		
Minimum	=	%
Maximum	=	%
Ramp Rate	=	%/s

MAIN, SETUP ➡

*** VDC ***		
Prim.NetID	=	
Sec.NetID	=	
Ramp Time	=	s

MAIN, SETUP ➡

** DIGITAL I/Os **		
I/Os		
Din from Ain		

Sub-menu 2

Sub-menu 3

Direction:	Input Din
	None, Excitation ON, Gen CB cl. st., Parallel wGrid, Increase, Decrease, Reset Setpoint, Remote SP Ena., PF Enable, Var Enable, Manual Enable, Open Loop Ena., Sync, VDC Enable, Secondary Net, Reset Alarm, Standby, Fieldbus Block, FCB Closed st.

Sub-menu 4

*** DIO1
*** DIO2
*** DIO3
*** DIO4 ***
Direction =
Din =
Dout =
Polarity =

** DIGITAL I/Os**
DIO1
DIO2
DIO3
DIO4
DI5
DI6
DI7
DI8

*** DI5
*** DI6
*** DI7
*** DI8 ***
Din =
Polarity =

Direction:	Output Dout
	None, Boost, Field Flashing, System OK, Limit Active, V/Hz Lim. Act., SP Lim. Reach., SP Min. Reach., SP Max. Reach., Oper.Lim.Act., Min Ie Act., Max Ie Act., Min PQ Act., Min UM Act., Max UM Act., Voltage Relay, Close CB Comd, Sync Check, SW Alarm, Diode Alarm, Diode Trip, SW or Di Trip, Fieldbus Alarm, FCB Close CMD, FCB Open CMD

Sub-menu 4

** DI09
** DI10
** DI11
** DI12
** DI13
** DI14 from -AI *
Din =
Polarity =

** DIN from AIN *
DI9 from +AI
DI10 from -AI
DI11 from +AI
DI12 from -AI
DI13 from +AI
DI14 from -AI

Polarity:

**Normal,
Inverted**

Sub-menu 2

Sub-menu 3

<i>Ain:</i>	<i>None, Auto Remote SP, PF Remote SP, Var Remote SP, Manual Rem SP, OpenLoop R SP, UM Aux, Ie Ext [protected], Cool Med Temp, Reserved4, Reserved5, DI9(+)&10(-), DI11(+)&12(-), DI13(+)&14(-)</i>
-------------	---



MAIN, SETUP ➡

*ANALOG INPUTS
AI1
AI2
AI3
UM Aux
Ie Ext [protected]

***	AI1	
***	AI2	
***	AI3	***
Ain =		
Uin0%	=	V
Uin100%	=	V

MAIN, SETUP

ANALOG INPUTS ➡

***	UM AUX	
***	Ie Ext	***
Minimum	=	A
Maximum	=	A

<i>Aout:</i>	<i>None, Exc. Current, PWM, Fbias</i>
--------------	---



MAIN, SETUP

ANALOG OUTPUTS
AO1
AO2
Ie Range
Fbias Range

***	AO1	
***	AO2	***
Aout =		
Uout0%	=	V
Uout100%	=	V

MAIN, SETUP ➡

ANALOG
OUTPUTS ➡

**	Ie RANGE	
*	Fbias RANGE	*
Fbias 0%	=	%
Fbias 100%	=	%

Sub-menu 1

Sub-menu 2

Sub-menu 3

MAIN, SETUP ➡

DIODE MONITORING		
f Exc Nom	=	Hz
Tconst Exc	=	ms
Active	=	
Alarm Lvl	=	%
Alarm Dly	=	s
Trip Level	=	%
Trip Delay	=	s

MAIN, SETUP ➡

PSS		
Active	=	FALSE
PSS Settings		
PSS Time Consts		

* PSS SETTINGS*		
KS1	=	pu
KS2	=	pu
KS3	=	pu
M	=	
N	=	
T7	=	s
PSS max	=	%
PSS min	=	%
P min	=	%

MAIN, SETUP

PSS ➡

* PSS TIME CONSTS		
TW1	=	s
TW2	=	s
TW3	=	s
TW4	=	s
T1	=	s
T2	=	s
T3	=	s
T4	=	s
T8	=	s
T9	=	s
T10	=	s
T11	=	s

MAIN ➡

*** TUNE ***		
Auto		
PF/Var/PQ Limiter		
Manual/le Limiter		
Open Loop		

*** AUTO ***		
Setpoint	=	%
Vp	=	
Ta	=	s
Tb	=	s
Kq	=	%
Step	=	%
Do Step		

Sub-menu 1

MAIN, TUNE ➡

Sub-menu 2

PF/Var/PQ LIMITER			
SP PF	=		
SP Var	=	%	
Vp	=		
Ta	=	s	
PF-Step	=		
Var-Step	=		
Do PF-Step			
Do Var-Step			

MAIN, TUNE ➡

MANUAL/le LIMITER			
Setpoint	=	%	
Vp	=		
Ta	=	s	
Step	=	%	
Do Step			

MAIN, TUNE ➡

** OPEN LOOP **			
Setpoint	=	%	
Step	=	%	
Do Step	=		

MAIN ➡

* COMMUNICATION	
AVR ID	=
Modbus Setup	

MODBUS SETUP	
Slave ID	=
Active	=
Bitrate	=
Framing	=
Protocol	=
Answ Delay	= ms
Keep Alive	= s
KpAlive Act	=

MAIN ➡

*** MAIN ***	
Write EEPROM	
-OK to write value	
-ESC to cancel	

4.1.5 Fault message on the display

In certain circumstances it could appear the following fault messages on the display:

- XIRQ, SI or SYSTEM HALTED

The regulator continues to work faultless and has not to be taken out of service. But, no parameters can be entered or changed.

The display is reset when the unit is taken out of service and the auxiliary power is disconnected for a short time.

A defective unit should be sent in for repair. See manufacturer's details on page 2.

4.2 PC-Software

The delivered PC software is called **CMT 1000** for UNITROL 1000-15.

System requirements:

- at least Pentium II (200MHz) processor
- 256 MB RAM
- 4-speed CD-ROM drive
- Microsoft® Windows™ 95/98, NT, 2000, XP

Installing CMT 1000 4.x software

To install the CMT 1000 software please carry out the following steps:

- Place the CMT 1000 CD in your CD-ROM drive
- Open the Windows File Manager or Explorer
- Display the CD root directory
- Double-click **Setup.exe** from the directory **CMT 1000 \ Installer**
- Follow the instructions that appear on the screen. The **Setup** loads the CMT 1000 program into your hard disk.



setup.exe

The application program is now installed.

To start the program:

- Open the user interface via Start button, Programs, UNITROL 1000-15 and Double-click the shortcut icon CMT 1000.



CMT 1000
v4.3xx.exe

No communication between CMT 1000 and PC:

If CMT 1000 does not communicate with the PC, it has to be checked whether the interface COM x is not adjusted or already occupied. See Port settings on page 65.

Port settings: Base I/O Port address: 3f8 (from windows control panel)
Interrupt Request Line (IRQ): 4

Distribution rights:

CMT 1000 was developed with the Lab VIEW software and is distributed free of license charges and costs. The copyright to Lab VIEW is owned by the company National Instruments International Distribution (NIID).

“Copyright © [02-03-20] National Instruments Corporation.”

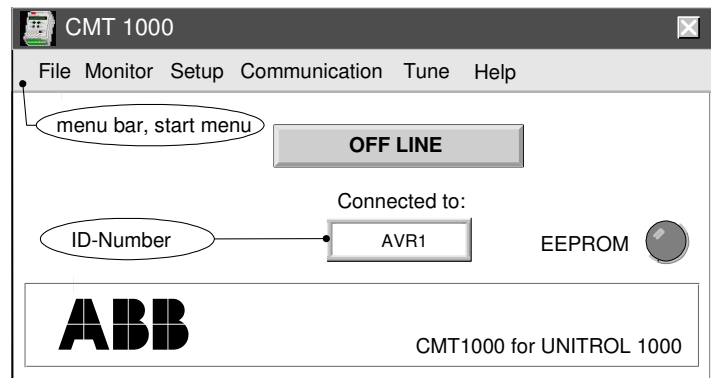
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4.2.1 Operation using the software

The window CMT 1000 is the graphic user interface of the program.

The other sub-menus are called up from the **menu bar**:

- File: Load/save parameters
- Monitor: Online measuring
- Setup: Set parameters
- Com.: Set ID number/ Port/ MODBUS
- Tune: Tune regulator
- Help: Software information



Display on the unit:

After the connection cable has been connected, the unit display reads *REMOTE CONTROL*.



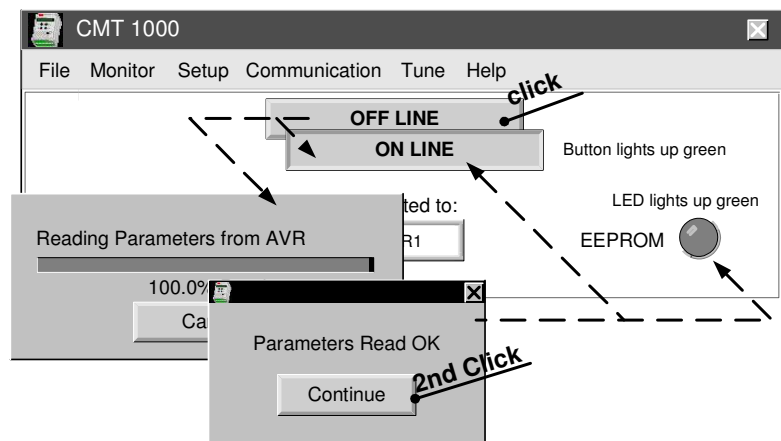
To make the communication between CMT 1000 and the unit:

The cable connection should be attached only after switching on the power of UNITROL 1000-15. It is possible to connect or disconnect the cable while the unit is in operation.

- Click the button OFF LINE

Caution: This button is shaded, if CMT 1000 cannot communicate with the unit.

- Press the button „Continue“ to accept the reading of the parameters.



The button is now with ON LINE labelled and lights up green
The parameters have been loaded from the unit to CMT 1000 (EEPROM LED lights up green)

4.2.2 Example of parameter setting using CMT 1000

Example: Assigning *Parallel with Grid Status* to input DI6
Make ON-LINE connection to the unit

Menu bar CMT 1000, start menu

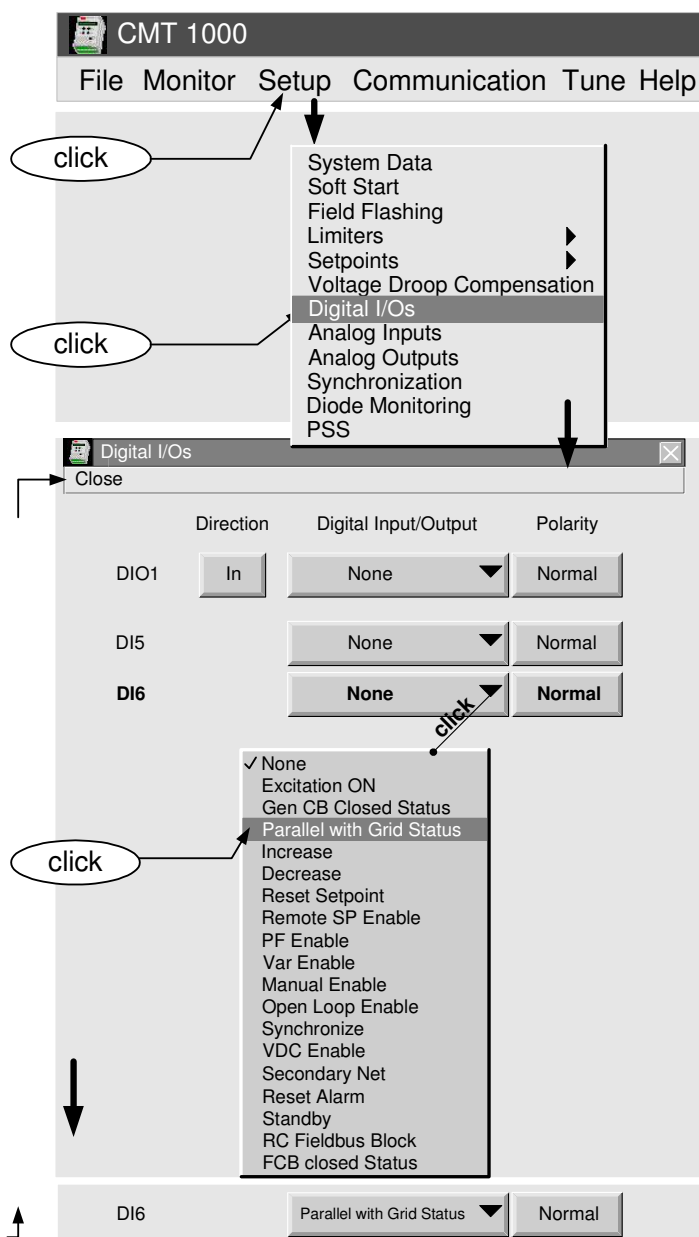
- Select setup

- Select digital I/Os

- Open input DI6 with ▼ menu

- Select input

Return to start menu



The above procedure only changes the parameters in the volatile memory.
To make the modification permanent, you have to save the entire parameter set.

Save parameters in ON LINE status:

- Unit: **Menu bar CMT 1000 \ File \ Write Parameters to EEPROM**
- Hard disk: **Menu bar CMT 1000 \ File \ Save Parameter File**

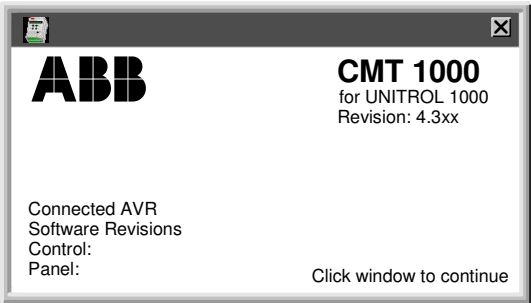
4.2.3 Menu structure of CMT 1000



About CMT 1000

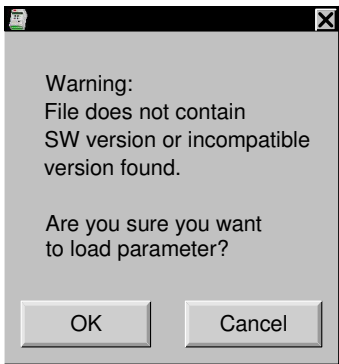
Control: Software revision unit
Panel: Software revision panel
Revision: Software revision CMT 1000

Click window to continue

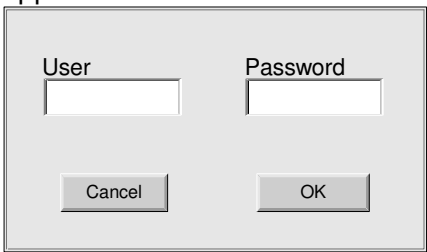


File	Monitor	Setup	Communication	Tune	Help
Open Parameter File					
Save Parameter File					
*.ini (see the warning message below) saves the parameters to the PC hard disk					
Write Parameters to EEPROM					
saves the parameters in the device's EEPROM					
Customer Setup					
user defined application, password detected					
Close					
closes the application					

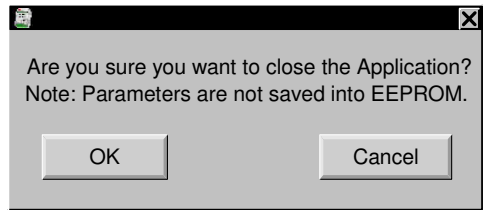
[Open Parameter File]

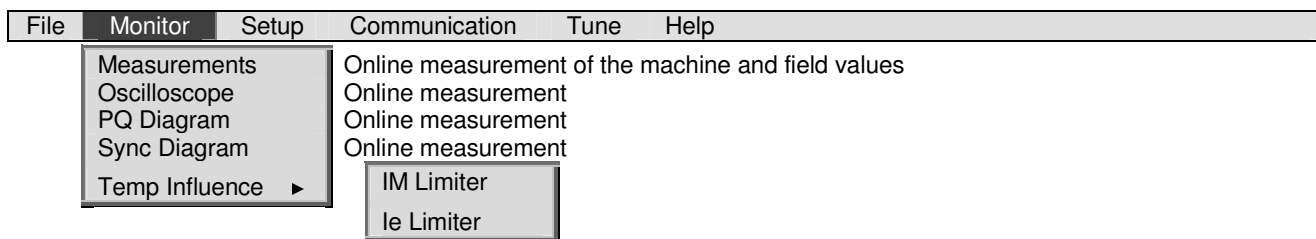


[Customer Setup]
Option for user defined application



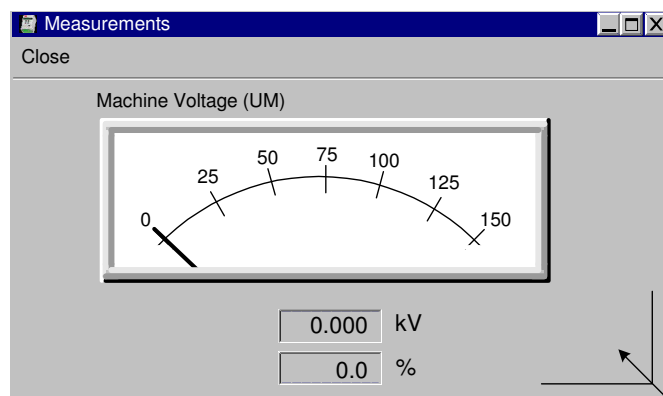
[Close]
See chapter 4.2.4, Save parameter file





- **Window measuring instruments**
Monitor \ Measurements

- | | |
|------------------------------|--------------|
| - Network Voltage (UNet) | [% and kV] |
| - Machine Voltage (UM) | [% and kV] |
| - Machine Real Power (P) | [% and kW] |
| - Excitation Current (Ie) | [A] |
| - Machine Current (IM2) | [% and A] |
| - Machine Reactive Power (Q) | [% and kVar] |
| - Network Frequency (fNet) | [Hz] |
| - Machine Frequency (fM) | [Hz] |
| - Power Factor (PF) | -- |



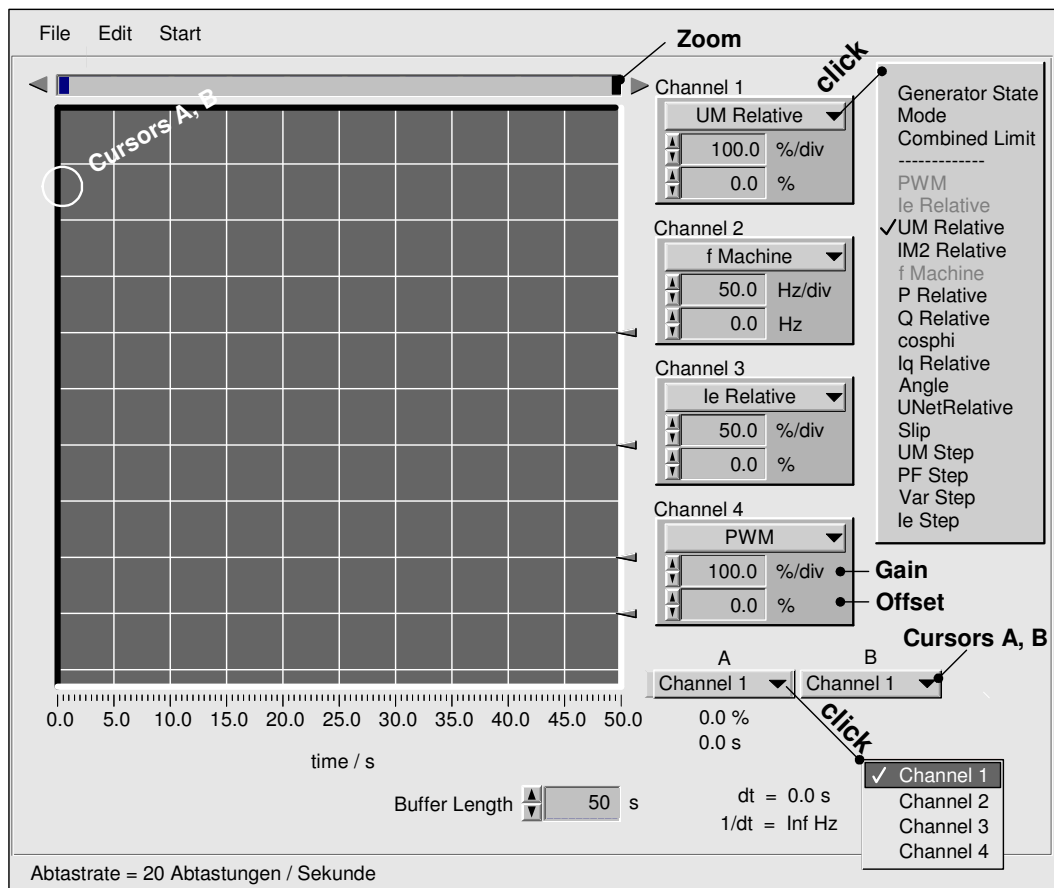
- **Oscilloscope**

Monitor \ Oscilloscope

File	Edit	Start
Open Waveform	Ctrl+O	open the waveform (is saved as ASCII table, *.xls)
Recall Setup		open settings of oscilloscope (*.cfg)
Save Waveform		save measurements as ASCII table (*.xls)
Save Setup		save oscilloscope settings (*.cfg)
Print Bitmap to File	Ctrl+B	save waveform in a BMP File
Close	Ctrl+Q	close application

File	Edit	Start
Cursors	Ctrl+C	show / hide cursors, Cursor appear on the left side of the screen
Sweep Buffer	Ctrl+S	Run through recording from Cursor A

File	Edit	Start
Start		start the record (appears menu point Freeze Waveform)
Freeze Waveform		stop the record (appears menu point Start)



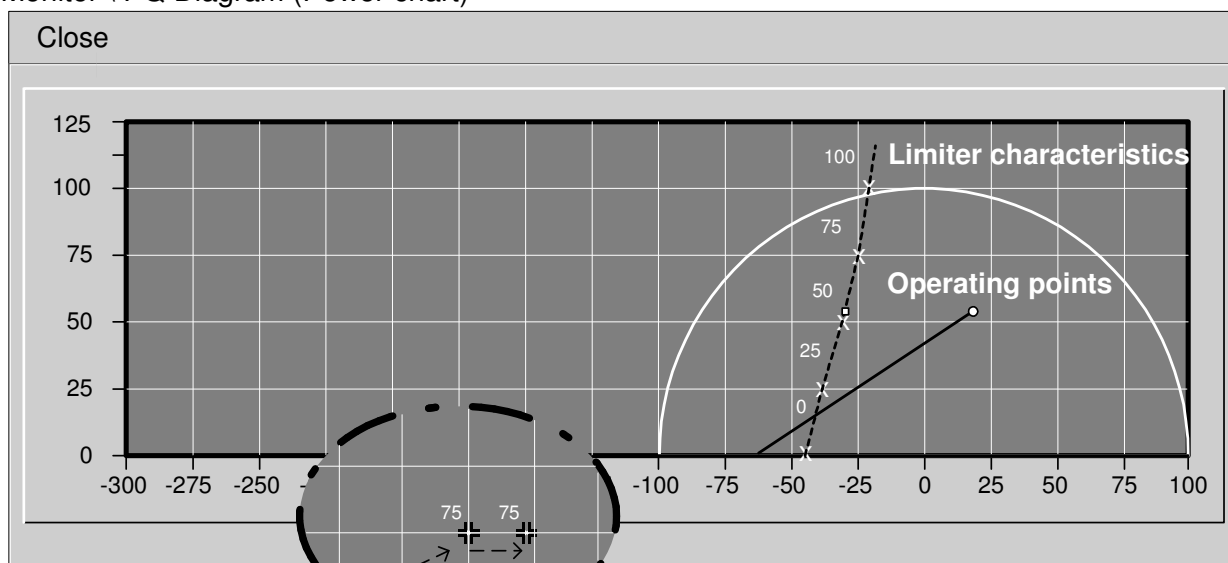
Measurement: Define buffer length (max. length = 100 s)

Start, stop and save record

The assignment of the signal, gain, offset can be chosen afterwards.

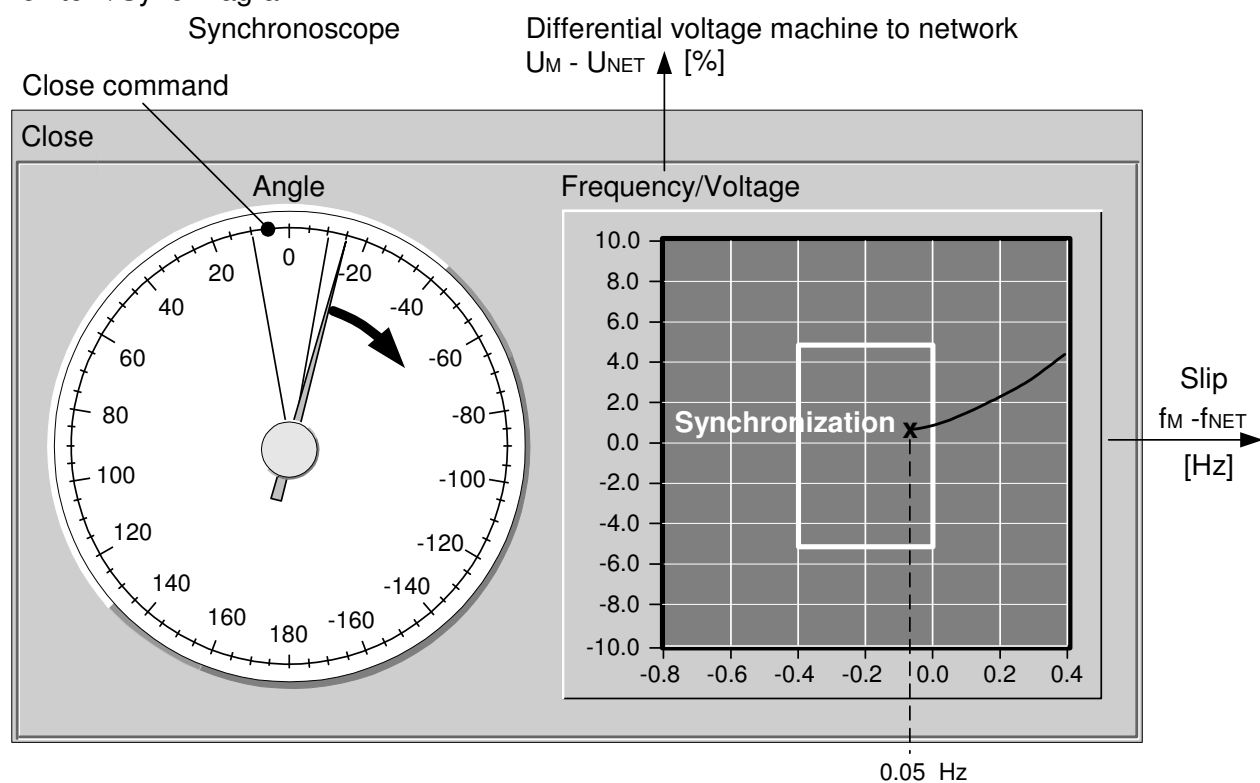
A reduced buffer length improves considerably the performance of the PC.

- **Window PQ Diagram**
Monitor \ PQ Diagram (Power chart)



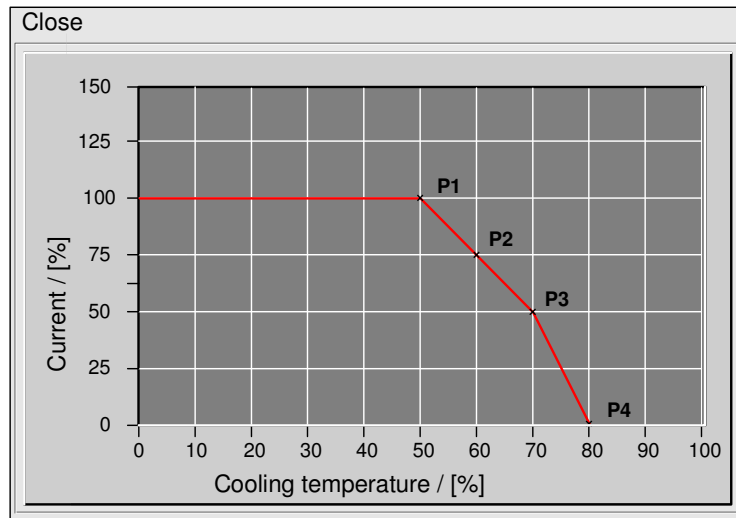
The Limiter values can be moved by cursor

- **Window Synchronization Diagram**
Monitor \ Sync Diagram

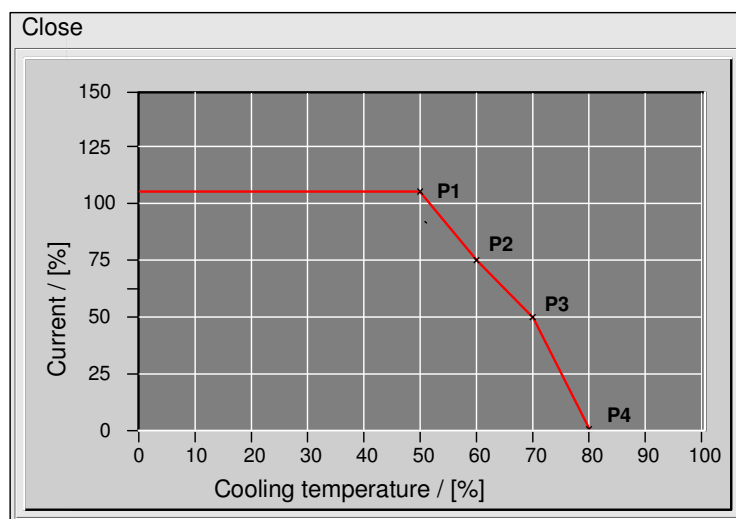


- **IM or Ie Temperature Monitor**
Monitor \ Temp Influence

IM Temperature Monitor



Ie Temperature Monitor



The Limiter values (Px) can be moved by cursor

- System Data
- Soft Start
- Field Flashing
- Limiters ▶
- Setpoints ▶
- Voltage Droop Compensation
- Digital I/Os
- Analog Inputs
- Analog Outputs
- Synchronization
- Diode Monitoring
- PSS

Setting of the system and unit parameters

- System Data
- Soft Start
- Field Flashing
- Limiters
- Setpoints
- Voltage Droop Compensation Control
- Digital I/Os
- Analog Inputs
- Analog Outputs
- Synchronization
- Diode Monitoring
- Power System Stabilizer

Limiters

- V/Hz Limiter
- Operational Limits
- Boost
- Temp Influence

PQ, I_e and UM limiter
Line short circuit support
IM and I_e temp. monitor

Setpoints

- Auto
- PF
- Var
- Manual
- Open Loop

Min., max. and ramp

- **Adjust V/Hz Limiter**

Setup \ Limiters \ V/Hz Limiter

Close	
f knee	48.0 Hz
Slope	100.0 %/fknee

- **Adjust Ie, UM, PQ and IM Limiter**

Setup \ Limiters \ Operational Limits \ Limiter Setup

Ie Limiter

Max

Close		
Max Ie	Max UM	Max IM
Min Ie	Min UM	Min Q
☒ Active		
Max Ie =	160.0 %	for 10.0 s
Max Delayed Ie =	120.0 %	for 60.0 s
Max Continuous Ie =	105.0 %	

Min

Close		
Max Ie	Max UM	Max IM
Min Ie	Min UM	Min Q
☐ Active		
Min Ie =	0.0 %	

UM Limiter

Max

Close		
Max Ie	Max UM	Max IM
Min Ie	Min UM	Min Q
☒ Active		
Max UM =	110.0 %	

Min

Close		
Max Ie	Max UM	Max IM
Min Ie	Min UM	Min Q
☒ Active		
Min UM =	90.0 %	

PQ Limiter

Close		
Max Ie	Max UM	Max IM
Min Ie	Min UM	Min Q
☒ Active ☒ Voltage dependency		
Min Q5 =	-20.0 %	when P = 100%
Min Q4 =	-25.0 %	when P = 75%
Min Q3 =	-30.0 %	when P = 50%
Min Q2 =	-35.0 %	when P = 25%
Min Q1 =	-40.0 %	when P = 0%

IM Limiter

Close		
Max Ie	Max UM	Max IM
Min Ie	Min UM	Min Q
☐ Active		
Max IM	100.0 %	
Time Multiplier K	1.00	

- **Adjust Short Circuit Support**

Setup \ Limiters \ Boost

Close

Threshold	40.0	%
Hold Time	3.0	s
Hysteresis	15.0	%

- **Adjust Limiters Temperature Influence**

Setup \ Limiters \ Temp Influence

Close

IM Limiter	IM2 Nom	500.0	A	le Limiter	le Nom	5.0	A				
T1	50.0	°	Max IM	100.0	%	T1	50.0	°	Max Cont le	105.0	%
T2	60.0	°	IM th2	75.0	%	T2	60.0	°	le th2	75.0	%
T3	70.0	°	IM th3	50.0	%	T3	70.0	°	le th3	50.0	%
T4	80.0	°		0.0	%	T4	80.0	°		0.0	%

IM and le can be changed only in “Operational Limits” on previous page.

- **Adjust Setpoint Range**
Setup \ Setpoints \ AUTO

Close

Minimum	90.0	%
Maximum	110.0	%
Ramp Rate	0.30	%/s

The setpoints *PF*, *Var*, *Manual* and *Open loop* are entered in identical input fields.

- **Adjust Voltage Droop Compensation Control Range**
Setup \ Voltage Droop Compensation

Close

Primary Net ID	1
Secondary Net ID	2
VDC Ramp Time	10.0 s

- **Configure Digital I/O**
Setup \ Digital I/Os

Close

	Direction	Digital Input/Output	Polarity
DIO1	In	None	Normal
DIO2	In	Out	Normal
DIO3	In	None	Normal
DIO4	In	None	Normal
DI5		None	Normal
DI6		None	Normal
DI7		None	Normal
DI8		None	Normal
DI9 from +AI		None	Normal
DI10 from -AI		None	Normal
DI11 from +AI		None	Normal
DI12 from -AI		None	Normal
DI13 from +AI		None	Normal
DI14 from -AI		None	Normal

Direction = In

✓None
 Excitation ON
 Gen CB Closed Status
 Parallel with Grid Status
 Increase
 Decrease
 Reset Setpoint
 Remote SP Enable
 PF Enable
 Var Enable
 Manual Enable
 Open Loop Enable
 Synchronize
 VDC Enable
 Secondary Net
 Reset Alarm
 Standby
 RC Fieldbus Block
 FCB closed Status

Direction = Out

✓None
 Boost
 Field Flashing
 System OK
 Limit Active
 V/Hz Limit Active
 SP Limit Reached
 SP Minimum Reached
 SP Maximum Reached
 Operational Limit Active
 Min Ie Active
 Max Ie Active
 Min PQ Active
 Min UM Active
 Max UM Active
 Voltage Relay
 Close CB Command
 Sync Check
 SW Alarm
 Diode Alarm
 Diode Trip
 SW Alarm or Diode Trip
 MODBUS Com Alarm
 Close FCB Command
 Open FCB Command

- **Configure Analog Inputs**

Setup \ Analog Inputs

Close

		min	max
	UM Aux	-10.0 %	10.0 %
	Analog Input		
AI1	None ▼	0.0 V	10.0 V
AI2	None ▼	0.0 V	10.0 V
AI3	None ▼	0.0 V	10.0 V

✓None
Auto Remote Setpoint
PF Remote Setpoint
Var Remote Setpoint
Manual Remote Setpoint
Open Loop Remote Setpoint
UM Aux
Reserved2
Cooling Media Temperature
Reserved4
Reserved5
Digital Input 9(+) & 10(-)
Digital Input 11(+) & 12(-)
Digital Input 13(+) & 14(-)

- **Configure Analog Outputs**

Setup \ Analog Outputs

Close

		min	max
	Ie Range	0.0 %	300.0 %
	PWM	0.0 %	100.0 %
	Fbias Range	-3.0 Hz	3.0 Hz
	Analog Output		
AO1	None ▼	0.0 V	10.0 V
AO2	None ▼	0.0 V	10.0 V

✓None
Excitation Current
PWM
Fbias

- **Adjust Synchronization**
Setup \ Synchronization

Close

Minimum Slip	0.00	Hz
Maximum Slip	-0.40	Hz
Maximum Delta U	5.00	%
Maximum Delta Angle	10	deg
Total CB Closing Time	90	ms
Voltage Offset	0.0	%
Angle Offset	0.0	deg

- **Adjust Diode Monitoring**
Setup \ Diode Monitoring

Close

☐ Active

Nominal Exciter Frequency	50	Hz
Exciter Time Constant	350	ms
Alarm Level	5.0	%
Alarm Delay	10.0	s
Trip Level	20.0	%
Trip Delay	0.3	s

- **Adjust PSS**
Setup \ PSS

Close

PSS_Select
Active ☐

Activation Level
P min %

Wash out time constants

Gains

TW1 s TW3 s
TW2 s TW4 s

PSS gain p.u.
factor KS1 p.u.
Compensation factor for
power integration KS2 p.u.
Signal matching
factor KS3 p.u.

Lead time constants
of conditioning network

Time constant
for power integration

Ramp tracking
filter time constants

T1 s
T3 s
T10 s

T7 s

T8 s
T9 s

Lag time constants
of conditioning network

Ramp tracking
filter degrees

Limits

T2 s
T4 s
T11 s

M
N

PSS max %
PSS min %

File	Monitor	Setup	Communication	Tune	Help
			ID Definition Port Configuration MODBUS	Set ID number Set Port number Set MODBUS configuration	

- **Set ID Number**

Communication \ ID Definition

Required in VDC mode to be set unique

Close

ID

- **Set Port Configuration**

Communication \ Port Configuration

Push the button to select one of the available COM-ports.

ASRL1 (COM1-Communications Port)
 ✓ ASRL1 (COM1-Communications Port)
 ASRL2 (COM2-Communications Port)
 ASRL10 (LTP1-Printer Port)

Cancel OK

CMT 1000 must be restarted
OK

- **MODBUS Configuration**

Close

Slave ID Answer delay ms Bit rate b/s Keep-alive time sec

Active ☐ Protocol Character framing Keep-alive action

✓ RTU TCP

✓ Even Odd none none

✓ none Set SW alarm Set fieldbus block Clear fieldbus request

Good messages

Bad messages

Respond counter

MODBUS register

Address Data

Integer Bit Hex

MODBUS register

Address Code Data

0 0 0

Integer Bit Hex

File
Monitor
Setup
Communication
Tune
Help

Setpoint Adjust
Auto
PF/Var/PQ Limiter
Manual/Ie Limiter

Automatic voltage regulation
PF, Var regulation and PQ Limiter
Manual control and Ie Limiter

- Adjust Setpoint, Trigger Setpoint, Status Displays
Tune \ Setpoint Adjust

Close

Generator State

Grid
Sec. Net
Prim. Net
NoLoad

Active Mode

Standby
Sync
Manual
Open Loop
Pf
Var
VDC
AUTO

Limiter State

Ie High
UM High
V/Hz
IM High
Ie Low
UM Low
Iq Low

Setpoint

100.0
95.0
105.0
90.0
110.0

DOWNUP

100.0 %

UM Step

0.0 %

DOWNUP

A setpoint jump can also be triggered in this window.

Adjust setpoint

Set step height

Trigger setpoint jump

- Parameter Setting Range
Tune \ Auto

Close

Vp
(Proportional Gain)

75
100
125
150
175
200
25
50

20.0

Ta
(Integration Time)

10.00
15.00
20.00
5.00
0.01

4.00 s

Kq
(Droop)

0
5
10
15
20
-5
-10
-15
-20

0.0 %

Tb
(Derivation Time)

2.00
3.00
4.00
1.00
0.01
5.00

0.20 s

- Tune \ PF/Var/PQ Limiter or
- Tune \ Manual/Ie Limiter

Close

Vp
(Proportional Gain)

75
100
125
150
175
200
25
50

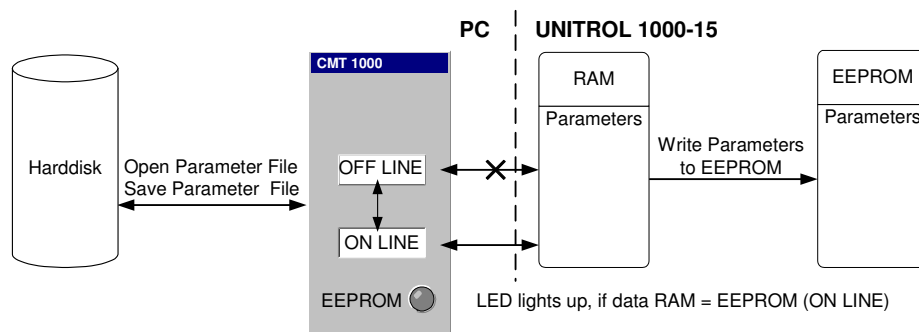
10.0

Ta
(Integration Time)

10.00
15.00
20.00
5.00
0.01

3.00 s

4.2.4 Save parameter file



- **Status ON-LINE**

After switching from OFF LINE to ON LINE, CMT 1000 reads the parameters from the device's RAM. Now, all parameter changes carried out using CMT 1000 are written directly into the device's RAM.

- Write Parameters to EEPROM: The parameters in the RAM are stored in the device.
- Save Parameter File: The parameters are saved onto the hard disk as an INI file.
- Open Parameter File: The INI file is loaded into CMT 1000 and the RAM is overwritten again.

When opening a parameter file from disk, the old parameter file is immediately overwritten in the RAM!

- **Status OFF-LINE**

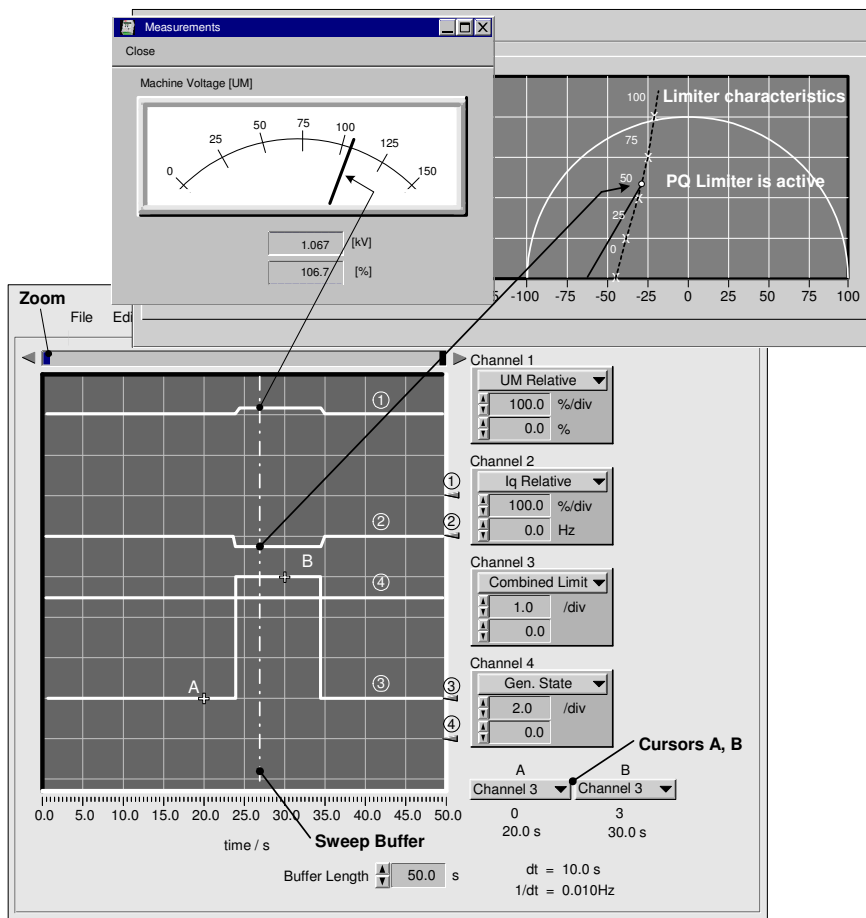
Any parameter changes carried out in CMT 1000 are not transferred into the device.

- Write Parameters to EEPROM: No effect, since there is no communication between CMT 1000 and the device.
- Save Parameter File: The parameters are saved onto the hard disk as an INI file.
- Open Parameter File: The INI file is only loaded into CMT 1000.

If you now switch to ON-LINE, CMT 1000 reads the old data from the RAM again!

4.2.5 Evaluation of the waveform

The curves for the current and stored data can be viewed using the oscilloscope, PQ diagram and the instruments. (Monitor toolbar)



- **Status OFF LINE**

File \ Open Waveform: Open stored waveform file.

Edit \ Cursors: Assign channel with buttons A and B, use left mouse button to drag cursor A to the desired point on the signal curve. The measured values can be measured on the oscilloscope, instruments or PQ diagram.

Edit \ Sweep Buffer: The curves are run through from cursor point A using the scanning bar.

- **Status ON LINE**

Start: The recording of the waveform is started.

Freeze Waveform: The recording is stopped.

File \ Save Waveform: Save waveform.

File \ Save Setup: Save oscilloscope settings.

- The window in “Tune” can also be opened to set the regulators.
- The setpoint step is simulated in “Tune Setpoint Adjust”.
- The buffer length can be adjusted from 1 s, 2 s, 5 s, 20 s, 50 s to 100 s.
- The time scale can be adjusted in the bar above the waveform (Zoom).

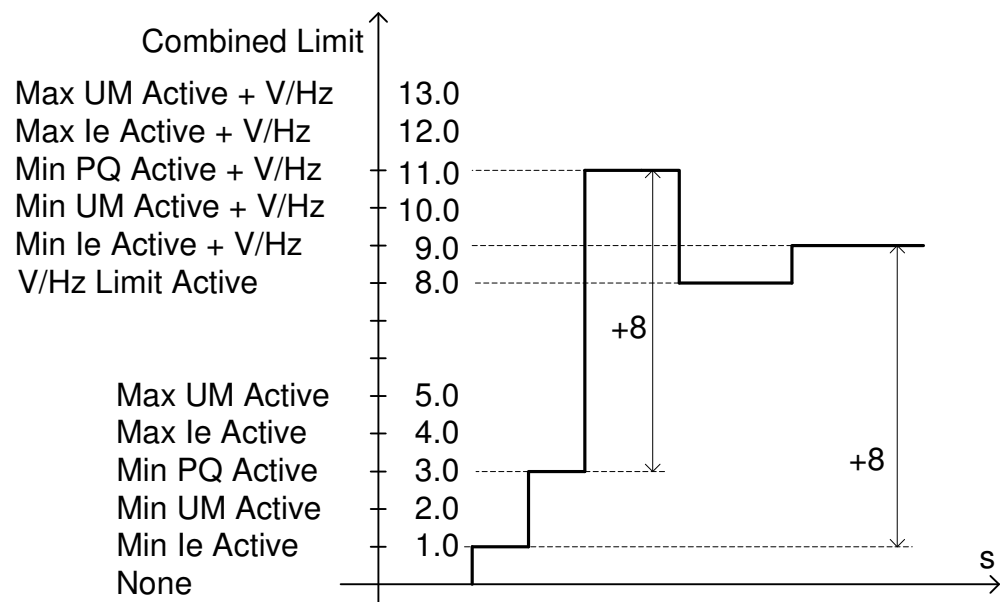
Monitor \ oscilloscope, **status display**

There are 3 quasi analog signals which can be used to represent status changes on the oscilloscope. Decoding of the signals *Generator State*, *Mode* and *Combined Limit*.

	signal value	state
Generator State Mode Combined Limit ----- PWM I _e Relative UM Relative IM2 Relative f Machine P Relative Q Relative cosφ _i I _q Relative Angle UNetRelative Slip UM Step PF Step Var Step I _e Step	Generator State:	0 = Idle (NoLoad)
	1 =	Change NoLoad -> Primary Net
	2 =	Primary Net
	3 =	Change Primary Net <-> Secondary Net
	4 =	Secondary Net
	5 =	Change Secondary Net -> NoLoad
	6 =	Primary Net or Secondary Net -> Grid ON
	7 =	Grid ON
	8 =	Grid ON -> Primary Net or Secondary Net
	Mode:	0 = Auto
	1 =	Var
	2 =	PF
	3 =	Manual
	4 =	Open Loop
	5 =	VDC
	6 =	Sync
	7 =	Standby
	Combined Limit:	0 = None
	1 =	Min I _e Active
	2 =	Min UM Active
	3 =	Min PQ Active
	4 =	Max I _e Active
	5 =	Max UM Active
	+8 =	V/Hz Limit Active*)
	+16 =	SP Minimum Reached
	+32 =	SP Maximum Reached
	+64 =	Diode Alarm
	+128 =	Diode Trip
	+256 =	Max IM active
	Excitation ON active	Combined Limit = See signal value
	Excitation ON not active	Combined Limit = -1.0

- *) Each time a limiter is tripped, the V/Hz-Limiter, outside of the Setpoint Range and or the Diode Monitoring can also occur and be displayed in the status signal of the limiters. All other collective signals only allow one display.

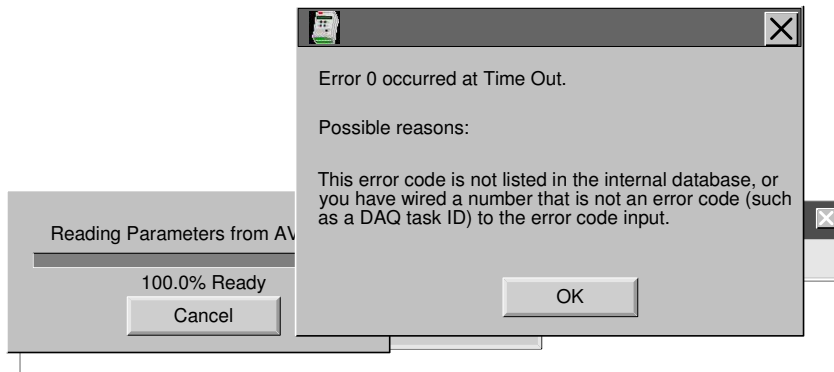
Summary of the limiter status:



4.2.6 Error message

- Error message during the reading of the parameters

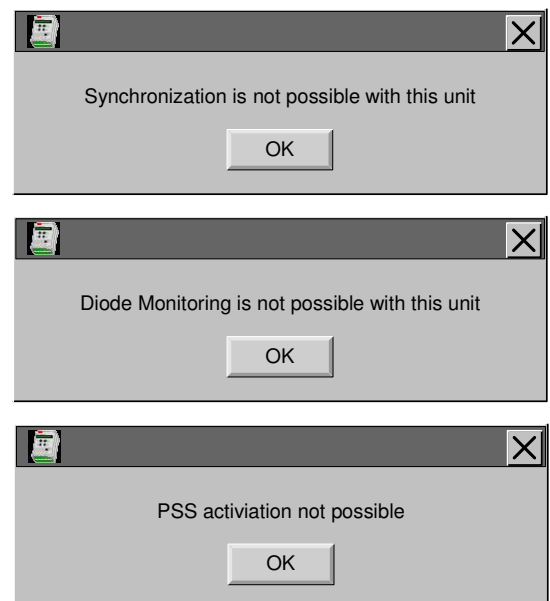
The following message appears if the connection is interrupted during the reading of the parameters.



- Acknowledge by clicking on the Continue button.
- Check connection between PC and device and click OFF LINE button again.

- Error message at assigning of the digital input with the signal *Synchronize*, *Diode Monitoring* respectively *PSS*.

The Error message appears, when the relevant unit does not include the option *Synchronization*, *Diode Monitoring* respectively *PSS*.



You can ask for information directly from the manufacturer, in case the option “Synchronization“, “Diode Monitoring” respectively “PSS” is required to this unit.

See manufacturer’s details on page 2

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5 COMMISSIONING

5.1 Safety regulations



WARNING!

UNITROL 1000-15 units operate with dangerous voltages of up to 250 VAC.

Manipulation of live parts can lead to death or injury to the persons involved or damage to the surroundings.

Possible risks are largely excluded if the unit is handled properly in accordance with these instructions.



WARNING!

The secondary voltage of the excitation transformer and the voltage of the excitation field are fed into the excitation cabinet.

These components present a great danger of electric shocks.

The control elements and the PC interface on the front plate of the UNITROL 1000-15 unit are to be touched and/or attached only under large caution.

After the unit has been switched off, it must be ensured by measurement that no measuring voltages or control voltages >50 V are present at the terminals. At an interrupted field circuit the input capacitor is slowly discharged through internal circuits.

In order to prevent unintentional closing of open voltage circuits by third parties, the circuits in question should be identified at the point of interruption (e.g. by means of a warning sign).

Before switching on, check whether the connection terminals are wired up according to the plant schematic.

5.2 Setting aids

At the first commissioning all parameters have to be controlled.

For this see description of the parameters in section 2.4.2 and settings recorded in section 7.3.

Overview of parameter blocks

- Setup:
 - System Data
 - Soft Start
 - Field Flashing
 - Limiters
 - Setpoints
 - Voltage Droop Compensation Control
 - Digital I/Os
 - Analog I/Os
 - Synchronization
 - Diode Monitoring
 - Power system stabilizer (PSS)
- Tune: Controller
- Communication: ID unit number/ Port/ MODBUS.

Setting aids

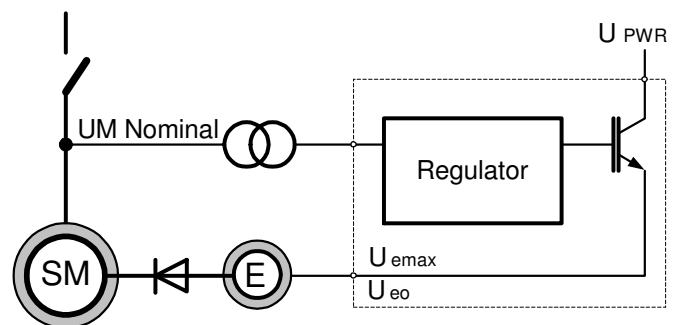
- Setup \ System Data, **Ceiling Factor Kceil**

The following section describes how to determine the Ceiling Factor in two different ways.
We recommend the measuring method.

Measuring method

$$K_{ceil} = \frac{U_{emax}}{U_{eo}} = \frac{100}{PWM_{noload}}$$

U_{emax}: maximum output voltage
 U_{eo}: output voltage at nominal voltage of the machine (machine in no-load operation)
 PWM no load: Regulator output at no load (PWM)



Definition: The ceiling factor K_{ceil} determines the relationship between the maximum output voltage of the regulator and the exciter voltage, which is required for no-load excitation and is thus a measure for the possible overexcitation of the machine.

Example: Regulator output signal (PWM) at no load shows 20% at nominal voltage

$$K_{ceil} = \frac{100}{20} = 5$$

Calculating method

Power electronics supply: three- phase: $U_{\max} = 1.40 \cdot U_{\text{PWR}} - 8 \text{ V}$
single-phase: $U_{\max} = 1.25 \cdot U_{\text{PWR}} - 8 \text{ V}$
DC: $U_{\max} = 0.98 \cdot U_{\text{PWR}} - 8 \text{ V}$

Example:

Specifications: Power electronics supply $U_{\text{PWR}} = 240 \text{ V}$, single-phase
No-load exciter voltage $U_{\text{eo}} = 50 \text{ V}$

Calculation: $U_{\max} = 1.25 \cdot 240 \text{ V} - 8 \text{ V} = 292 \text{ V}$
 $K_{\text{ceil}} = 292 \text{ V} / 50 \text{ V} = 5.8 \text{ p.u.}$

- Setup \ System Data, **Machine Reactance X_q (non-saturated)**
 - Salient pole machine: $X_q \approx 0.7 \text{ to } 0.5 X_d$
 - Cylindrical-rotor machine (Turbo): $X_q \approx X_d$
- Setup \ Limiter, **Manual and Open Loop**
There are no limiters active in the mode Manual and Open loop.
- Setup \ Limiter, **UM Limiter**
The limiting values of the UM limiter are, as a rule, set equally, like the setpoint range of the voltage regulator (Auto mode).

Setpoint Auto	UM Limiter	Example
Minimum	= Minimum	90%
Maximum	= Maximum	110%

- Setup \ **Digital I/Os, configuration**
Software configuration of the **digital I/Os**:
 - Check switching threshold of the inputs. Default values apply to internal 24 V supply.
 - Define terminal as input or output
 - Assign terminal to desired signal
 - Select polarity
- Setup \ **Analog inputs / outputs, configuration**
Software configuration of the **analog inputs**:
 - Assign terminal to desired signal
 - Define signal level of the input signal [U_{in} 0% to U_{in} 100%]
 - If the input is used as a digital input, the desired signal name is to be assigned in the menu

Digital I/Os.

Software configuration of the **analog outputs**:

- Assign terminal to desired signal
- Define signal level of the output signal [U_{out} 0% to U_{out} 100%]
- Define signal level of the field current [I_{e} 0% to I_{e} 100%]

- Setup \ Digital I/Os, **selection of operating mode**

The operating mode is selected via the digital inputs. For this purpose, a maximum of 10 inputs have to be occupied by the following signals:

Mode	Digital inputs which are to be assigned								
	Stand by	SYN	Gen CB Closed Status	Parallel with Grid Status	Manual Enable	Open Loop Enable	VDC Enable	PF Enable	Var Enable
Standby	1	X	X	X	X	X	X	X	X
Sync	0	1	X	X	X	X	X	X	X
Manual	0	0	X	X	1	X	X	X	X
Open Loop	0	0	X	X	0	1	X	X	X
VDC	0	0	1	0	0	0	1	X	X
PF	0	0	1	1	0	0	0	1	X
Var	0	0	1	1	0	0	0	0	1
Auto	0	0	0	X	0	0	X	X	X
with *)	0	0	1	X	0	0	0	0	0

*) with droop or compensation

0 = logical 0 or open

1 = logical 1,

X = not relevant

Note: If an input is to be continuously logical 1, then it can also be inverted by software means. The input then naturally counts as being occupied.

The inputs DI9 to DI14 can be used as virtual digital inputs even if they are not defined in Analog Inputs as Digital Inputs.

- Setup \ **Synchronization**

The machine frequency has to be always greater than network frequency. It cannot be synchronized by a positive slip.

- Setup \ **Diode Monitoring**

The Nominal and the Nominal Exciter Frequency have to be adjusted, for the other keep the default value.

Nominal Exciter Frequency = $p \times n / 60$

p = Number of poles pairs [.]

n = Revolutions per minute [U/min]

- Setup \ **PSS**

The parameters can be calculated by ABB Switzerland. For further information please see the Functional Description of UNITROL 1000, Power System Stabilizer, 3BHS213239 E01.

See manufacturer's details on page 2.

- Tune \ Setpoint Adjust, **Step**

Mode	Maximum adjustable setpoint jump during 10 s
Manual	±50%
Open Loop	±20%
PF	±0.2
Var	±50%
Auto	±20%

- Tune, **Order of priorities**

If several operating modes are selected simultaneously by the digital inputs, then the following order of priority applies. This is also shown in the tables *Selection of operating mode* on the preceding page.

Priority	Mode	Remark
1. resp. top	Standby	
2.	Sync	
3.	Manual	
4.	Open Loop	
5.	VDC	Only active, if <i>Gen CB Closed</i> status is active and <i>Parallel with Grid</i> status is not active
6.	PF	Only active, if <i>Parallel with Grid</i> status is active
7.	Var	Only active, if <i>Parallel with Grid</i> status is active
8. resp. lowest	Auto	Active, if no other operation mode is active

- Tune, **Auto**

The following settings can be pre-set on the basis of the machine data.

Integral time constant Ta

This lies within the range of Td' of the main machine.

Setting: $T_a = T_d'$ (Typically value 2 to 5 s)

Derivative time constant Tb

This approximately compensates the load time constant TE of the exciter machine.

Setting: $T_b = 0.7 \text{ to } 1.0 \cdot T_E$ (Typically value 0.1 to 0.3 s)

Proportional gain Vp

This most important setting value depends primarily on the controlled system.

If the ceiling factor Kceil has been set correctly, values between 10 and 40 should result in stable regulation of the machine.

A default value of 20 is set for the first excitation.

Reactive power influence Kq

In the case of machines which, without step-up transformers, are connected with the fixed grid or, via a busbar, with other synchronous machines, it is essential that the droop Kq is set to a **negative value** of -5 to -20%. A higher negative value makes the machine more independent of

voltage fluctuations in the grid; the reactive power output or consumption stabilises. At low negative values, the machine will support the grid or busbar voltage.
In the case of machines with step-up transformers, the voltage drop of the transformer can be partly compensated with positive values of Kq.

Example: Step-up transformer Reactance = 12%
 Compensation Kq = +6%

- Communication \ **MODBUS**

The RS485 port is shared between VDC mode and MODBUS. With factory settings, MODBUS is disabled.

To use MODBUS, the parameter MB_Enable must be set true, then the parameters have to be written to the EEPROM. Afterwards, the UNITROL 1000 must restart by power cycling. This procedure has to be followed for any change to MODBUS related parameters to take effect. It serves as a protection against inadvertently overwriting the setup through MODBUS itself and therefore loosing connection to the UNITROL 1000.

- Standard configuration for:
 - Answer Delay: 10 ms
 - Bit rate = 19200
 - Character Framing = even
 - RTU: protocol of the RS485

- Slave ID:

Zero and values greater 247 are reserved, so the valid range for the slave ID is 1 to 247, with 247 as default.

- Keep-alive time:

The parameter "MB_KeepAliveTime" defines, in which period of time the "Keep Alive Bit" in control word 1 must be changed. Valid settings are 1 to 120 seconds and 0 to disable the connection monitoring.

- Keep-alive action:

If the timer reaches the supervision time a "Modbus Communication Alarm" will appear and one of the following event can be selected with the "MB_KeepAliveAction".

For more information, refer to Modbus Manual

5.3 Work carried out while machine is at a standstill

Preliminary checks:

- Wiring check, compare connections with schematic



CAUTION!

Check of the insulation strength of the plant with the insulation tester

During the test the equipment could damage by the testing voltage.

Disconnect cables to the UNITROL 1000-15.

- Switch on auxiliary voltage UAUX
- Adjust parameters
- Check measuring voltage and current transformer circuits
- Measure field resistance
- Low-load test: External power electronics supply (3x <250 V AC)
 - Open loop mode
 - Optimise regulator Ie
- Adjust limiter settings on the basis of customer's power chart

5.4 Work carried out while machine is running

No-load tests: (nominal speed, not synchronized)

- Field flashing and discharge in Auto and Manual mode
- Soft start in Auto mode
- Adapt setpoint range, optimise voltage regulator
- V/Hz limiter: The actuation point fknee is set to 48 Hz by default

For 60 Hz machines this value must be set to approximately 58 Hz.

Tests under load:

- Stator current measurement, internal P- and Q-measurement, droop/compensation

The droop/compensation is set to 0% by default. If the machine is paralleled directly with the rigid grid or, via a busbar, with other synchronous machines, it is essential that the droop K_q is set to a value of approx. –10% before the first synchronization. The reactive current must be monitored carefully during the first paralleling. If it increases uncontrollably following the first synchronization, then the current polarity of the IM2 measurement is incorrect or the CT phase position is wrong.

- Load rejection overexcited and underexcited
- Optimise I_{emin} / I_{emax} Limiter
- Optimise PQ Limiter

The setting of the PQ limiter must be co-ordinated with the settings of the generator protection. As a rule, the limiter should be set at least 5% lower.

- PF and Var regulator, stability, setpoint range.

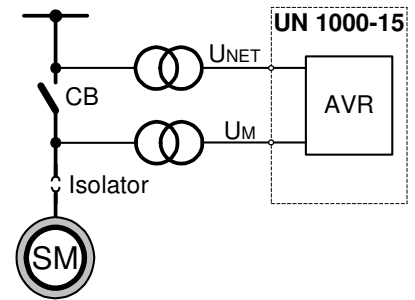
To enable correct operation of PQ Limiter, the PF / Var regulator must be tuned, even if these modes are not used.

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Synchronization:

It is a prerequisite for the correct functioning of the synchronization that there is no error in phase or magnitude between U_{NET} and U_M , caused by wrong connection or ratio of PT's on the grid side and generator side.

- Measurement of the phase angle of the voltages U_{NET} and U_M :
Open the isolator, close the circuit breaker (CB)
For reasons of safety has to be measured not only in the menu *Monitor\Oscilloscope* or *Monitor\Sync Diagram* but also with an oscilloscope.
- „Blind“ synchronization:
Watch the event with synchronoscope (*Monitor\Sync Diagram*) by disconnected trigger of the circuit breaker.
- „Live“ synchronization:
Record the event with synchronoscope
- About the measurement:



CAUTION!

In this status, the phase-angle difference must be 0 deg. Otherwise the measuring cable are incorrectly connected. Incorrect polarity cannot be detected by the UNITROL 1000-15.

Paralleling with incorrect polarity can lead to serious damage.

There must not occur a phase shift caused by one of the both measuring transformer or by the connection group of a step-up transformer.

The synchronoscope must stand at „twelve o'clock “. If the synchronoscope stands at „six o'clock“, one voltage is incorrectly connected. In other positions, either measuring is being carried out on incorrect phases or a phase shift caused by the connection group of a step-up transformer has not been compensated.

Concluding work:

- Saving the parameter and write the parameter settings section 7.3 or printing out the INI file.
The INI file can be opened using an editor (Word, Note- or WordPad) and printed out.

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6 MAINTENANCE AND FAULTS

6.1 Safety regulations



WARNING!

The secondary voltage of the excitation transformer and the voltage of the excitation field are fed into the excitation cabinet.

These components present a great danger of electric shocks.

The control elements and the PC interface on the front plate of the UNITROL 1000-15 unit are to be touched and/or attached only under large caution.

6.2 Maintenance

When the system is at a standstill, the screwed terminals, which can loosen due to vibrations should be checked for tightness. Dusty cooling flanges should also be cleaned.

6.3 Trouble shooting

The following instructions are intended to assist in localising a fault within the excitation system as a whole. However, it is not possible to deal with all eventualities in full.

List of possible faults

Possible causes	Checks, action
Machine is not excited	
- Field circuit interrupted - Field circuit-breaker doesn't close	- Check wiring for break - Check field circuit-breaker
<u>Shunt supply:</u> Field flashing does not work	Check field flashing circuit
No electronic supply UAUX	- Measure auxiliary supply UAUX - Check for tripped protective circuit-breaker
No control element supply UPWR	- Measure power electronics supply UPWR - Check for tripped protective circuit-breaker
<u>Shunt supply:</u> - Machine is only excited to the value supplied by the field flashing - Machine is first excited then discharges again	- Measure auxiliary supply UAUX, Measure power electronics supply UPWR - Check field flashing Off level - Check operating mode. For field flashing normally Auto is used - Check field flashing Off level - Check setpoint - Measure power electronics supply UPWR - If all supplies and setpoints are correct change unit
Setpoint error	- Check operating mode. For field flashing normally Auto is used - Check setpoint
Regulator error	- Measure output voltage at terminals 8, 9 (<i>Osc.</i>) -> positive - Measure voltage UPWM (<i>CMT 1000, Monitor</i>) -> positive rising

Possible causes	Checks, action
Overvoltage during build-up	
- Overvoltage caused by voltage regulator - Field flashing current too high	- Machine voltage UM present - Check system data - Check operating mode. For field flashing normally Auto is used - Check setpoint - Check thresholds of UM Limiter - Check regulator settings - Check design of the field flashing circuit. Field flashing should deliver 10 to 15% of the no load excitation current

Machine voltage not stable in no-load operation	
Regulator error	- Check operating mode. For field flashing normally Auto is used - Check setpoints - Check parameters of Auto regulator
Setpoint error	- Higher, lower inputs unstable - Externally input setpoint unstable
Control element fault	Check wiring, loose contact UM, Ie

Parallel operation with grid unstable. Periodic oscillation of reactive and possibly active power	
Regulator settings incorrect	Were changes made to the grid configuration? Additional outputs, loads etc. installed? - Yes: re-set regulator - No: check parameters of Auto and PF, Var regulator

Possible causes	Checks, action
Irregular instability, i.e. sporadic over- or underexcitation which is not caused by grid	
• Droop influence of the voltage regulator ineffective or IM2 measurement defective	– Check droop/compensation setting – Check external current transformer circuit – Gen CB Closed Status not active
• Machine within inadmissible operating range (normally protected by limiters)	– Bring machine into normal operating range by adjusting the setpoint. – Check setting of limiters
• Regulator fault	Measure output voltage at terminals 8, 9 (<i>oscilloscope</i>) Measure voltage UPWM (<i>CMT 1000, Monitor</i>) – Signals in phase opposition: unit defective – Signals in phase: disturbance possibly caused by the driving side of the machine or by disturbances from the grid

Operating point cannot be adjusted	
• Setpoint error	– Check operating mode. For field flashing normally Auto is used – Check setpoint
• Limiter active	– Bring machine into normal operating range by adjusting the setpoint. – Check setting of limiters
<u>Excitation with compounding and boost circuit:</u> Excitation is only supplied by the series compounding • No control element voltage • Regulator fault	– Measure power electronics supply UPWR – Check for tripped protective circuit-breaker – Check operating mode. For field flashing normally Auto is used – Check setpoint – Check parameters of Auto regulator

Possible causes	Checks, action
External controls faulty	
No external control voltage	– Measure control voltage – Check wiring
No internal control voltages Vdig, Vref	– Measure internal control voltages
Configuration of the digital or analog inputs or outputs is not correct	– Check configuration

6.4 Repair

The unit may not be opened. A defective unit should be sent in for repair with failure description and if possible, trending of failure. See manufacturer's details on page 2.

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7 APPENDIX

7.1 General Data

Ordering information

Device designation	UNITROL 1000-15, V4
Order number	Basic equipment with Synchronization with RDM with Synchronization and RDM without PSS with PSS with MODBUS
	3BHE014557 R00x3 3BHE014557 R00x4 3BHE014557 R00x5 3BHE014557 R00x6 3BHE014557 R000x 3BHE014557 R001x 3BHE014557 R002x

Mechanical data

Weight	5 kg
Protection class	IP20
Dimensions (HxWxD)	245x190x142 mm

Climatic stability

Temperature range for operation	0 to 60 °C
Temperature range for storage	-20 to +75 °C
Vibration	2 mm f<15 Hz, 0.7 g f>15 Hz
Shock	5 g, 11 ms pulse

Electrical data

Auxiliary supply UAUX:	Maximum power consumption	25 W
Power electronics supply UPWR:	Frequency	DC, 40 to 600 Hz
Excitation output Im2:	- Maximum continuous current	15 A
	- Current reduction for ambient temperatures >50 °C	1 A /degree
	- Overload (maximum 10 s)	30 A DC
	- Overload (maximum 4 min)	20 A DC
Frequency range of measuring values Um and Im2		10 to 100 Hz
Accuracy:	Voltage regulation	<0.1%

Test voltage:

Power electronics supply UPWR against case and auxiliary supply UAUX	2830 V DC, 2 s
Auxiliary supply UAUX against case and power electronics supply UPWR	2830 V DC, 2 s
Voltage measurement inputs UM, UNET:	High impedance voltage input without galvanic separation

According to (USA, Canada) UL specification:

The screw-type terminals for inputs and outputs are approved for maximum voltage	150 V
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Relevant standards, CE conformity

EMC directive:	89/336/EEC
Generic emission standard	IEC/EN 61000-6-4
Generic immunity standard	IEC/EN 61000-6-2

Ordering information connection cable PC to UNITROL 1000-15

Series standard cable (terminals 2 and 3 crossed)	DB9 f/f/ 3 m
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7.2 Settings record for UNITROL 1000-15

Name and address of customer:	_____

Plant:	_____
Order no.:	_____
Plant schematic no.:	_____

Device identification:

Type plate:	_____
Delivery date:	_____
Software revision:	- Control: _____
	- Panel: _____
	- CMT 1000: _____

Remarks:

Place and date of commissioning:

Name:	Company:
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7.3 Parameter settings, default values

Parameter	Parameter INI-File	Standard value	Setting range	Set	Unit
System Data	[SYSTEM DATA]				
- Nominal excitation current - Measuring volt., three- or single-phase	Ie Nominal Potential Transformer	5.0 Three_ Phase	0.1 to 1000.0 Three_Phase / Single_Phase / Three_ph_gnd		A
- Nominal voltage of the machine	UM Nominal	1.000	0.010 to 32.000		kV
- Potential transformer, primary voltage	UM Primary	1.000	0.010 to 32.000		kV
- Pot. transformer, secondary volt. 3-Ph	UM Secondary	110.0	60.0 to 250.0		V
1-Ph			60.0 to 150.0		V
3-Ph gnd			60.0 to 150.0		V
- Nominal machine current	IM2 Nominal	500	1 to 32000		A
- Current transformer primary current	IM2 Primary	500	1 to 32000		A
- Current transformer secondary current	IM2 Secondary	1.000	0.500 to 1.500		A
- No-load current	Ie No Load	30.0	0.1 to 100.0		%
- Ceiling factor	Kceiling	3.00	1.00 to 100.00		V/V
- Machine reactance	Xq	0.80	0.20 to 4.00		
- Frequency nominal	f Nominal	50.00	10.00 to 200.00		Hz
Field Flashing	[FIELD FLASHING]				
- Reached voltage off level	Off Level	0.0	0 to 95.5		%
Soft Start	[SOFT START]				
- Starting voltage	Starting Level	0.0	0 to 100.0		%
- Delay until ramp	Hold Time	0.0	0 to 327.0		s
- Ramp time	Ramp Time	10.0	0 to 327.0		s
Limiter					
V/Hz Limiter:	[V/Hz LIMITER]				
- Cut-off frequency	fKnee	48.0	10.0 to 200.0		Hz
- Slope	Slope	100.0	0 to 327.0		%
IeMin Limiter:	[IeMin LIMITER]				
- Minimum limit	Minimum	0.0	0 to 150.0		%
- Limiter active	Active	FALSE	TRUE / FALSE		
IeMax Limiter:	[IeMax LIMITER]				
- 1 st limit	Maximum	160.0	0 to 400.0		%
- Delay time	Maximum Hold Time	10.0	0 to 327.0		s
- 2 nd limit	Delayed	120.0	0 to 300.0		%
- Delay time	Delayed Hold Time	60.0	0 to 327.0		s
- Continuous current	Continuous	105.0	0 to 150.0		%
- Limiter active	Active	TRUE	TRUE / FALSE		
PQ Limiter:	[PQ LIMITER]				
- Q Limiter at P = 0%	Minimum Q (P @ 0%)	-40.0	-400.0 to -0.5		%
- Q Limiter at P = 25%	Minimum Q (P @ 25%)	-35.0	-400.0 to 0		%
- Q Limiter at P = 50%	Minimum Q (P @ 50%)	-30.0	-400.0 to 0		%
- Q Limiter at P = 75%	Minimum Q (P @ 75%)	-25.0	-400.0 to 0		%
- Q Limiter at P = 100%	Minimum Q (P @100%)	-20.0	-400.0 to 0		%
- Voltage dependency active	Voltage Dependency	TRUE	TRUE / FALSE		
- Limiter active	Active	TRUE	TRUE / FALSE		

Parameter	Parameter INI-File	Standard value	Setting range	Set	Unit
UM Limiter: - Limiting value mini. machine voltage - Limiting value maxi. machine voltage - Limiter low level - Limiter high level	[UM LIMITER] Minimum Maximum Minimum Active Maximum Active	90.0 110.0 TRUE TRUE	0 to 150.0 0 to 150.0 TRUE / FALSE TRUE / FALSE		% %
Boost: - Minimum machine voltage - Time boost output is active - Hysteresis	[BOOST] Threshold Hold Time Hysteresis	40.0 3.0 15.0	0 to 100.0 0 to 32.0 0 to 100.0		% s %
Range of Setpoints					
Auto: - Minimum value - Maximum value - Ramp rate	[AUTO SETPOINT] Minimum Maximum Ramp Rate	90.0 110.0 0.30	0 to 150.0 0 to 150.0 0.01 to 100.00		% % %/s
PF: - Minimum value - Maximum value - Ramp rate	[PF SETPOINT] Minimum Maximum Ramp Rate	-0.8500 0.8500 0.0050	-0.2500 to +0.2500 -0.2500 to +0.2500 0.0001 to 0.1000		/s
Var: - Minimum value - Maximum value - Ramp rate	[Var SETPOINT] Minimum Maximum Ramp Rate	0,0 100.0 1.00	-200.0 to +200.0 -200.0 to +200.0 0.02 to 100.00		% % %/s
Manual: - Minimum value - Maximum value - Ramp rate	[MANUAL SETPOINT] Minimum Maximum Ramp Rate	0.0 150.0 1.67	0 to 50.0 10.0 to 300.0 0.01 to 100.0		% % %/s
Open Loop: - Minimum value - Maximum value - Ramp rate	[OPEN LOOP SETPOINT] Minimum Maximum Ramp Rate	0.0 100.0 1.00	0 to 100.0 10.0 to 100.0 0.01 to 100.00	-- -- --	% % %/s
Voltage Droop Compensation Control					
- ID-Nr. Primary net - ID-Nr. Secondary net - Ramp time	Primary Net ID Secondary Net ID Ramp Up Time	1 2 10.0	1 to 31 1 to 31 0.1 to 60.0		s

Parameter	Parameter INI-File	Standard value	Setting range	Set	Unit
Digital I/Os					
DIO1: - Input - Polarity - Output - Direction	[DIGITAL I/O 1] Input Polarity Output Direction	None Normal None In	Signal name Normal / Inverted Signal name In / Out		
DIO2: - Input - Polarity - Output - Direction	[DIGITAL I/O 2] Input Polarity Output Direction	None Normal None In	Signal name Normal / Inverted Signal name In / Out		
DIO3: - Input - Polarity - Output - Direction	[DIGITAL I/O 3] Input Polarity Output Direction	None Normal None In	Signal name Normal / Inverted Signal name In / Out		
DIO4: - Input - Polarity - Output - Direction	[DIGITAL I/O 4] Input Polarity Output Direction	None Normal None In	Signal name Normal / Inverted Signal name In / Out		
DI5: - Input - Polarity	[DIGITAL INPUT 5] Input Polarity	None Normal	Signal name Normal / Inverted		
DI6: - Input - Polarity	[DIGITAL INPUT 6] Input Polarity	None Normal	Signal name Normal / Inverted		
DI7: - Input - Polarity	[DIGITAL INPUT 7] Input Polarity	None Normal	Signal name Normal / Inverted		
DI8: - Input - Polarity	[DIGITAL INPUT 8] Input Polarity	None Normal	Signal name Normal / Inverted		

Parameter	Parameter INI-File	Standard value	Setting range	Set	Unit
Assign analog input to digital signal					
DI9 from +AI: - Input - Polarity	[DIGITAL INPUT 9 from +AI] Input Polarity	None Normal	Signal name Normal / Inverted		
DI10 from -AI: - Input - Polarity	[DIGITAL INPUT 10 from -AI] Input Polarity	None Normal	Signal name Normal / Inverted		
DI11 from +AI: - Input - Polarity	[DIGITAL INPUT 11 from +AI] Input Polarity	None Normal	Signal name Normal / Inverted		
DI12 from -AI: - Input - Polarity	[DIGITAL INPUT 12 from -AI] Input Polarity	None Normal	Signal name Normal / Inverted		
DI13 from +AI: - Input - Polarity	[DIGITAL INPUT 13 from +AI] Input Polarity	None Normal	Signal name Normal / Inverted		
DI14 from -AI: - Input - Polarity	[DIGITAL INPUT 14 from -AI] Polarity Polarity	None Normal	Signal name Normal / Inverted		

Signal name:

Digital inputs

- DIO1 to 4,
- DI5 to 8,
- (DI9 to14)

Direction = In

✓None
 Excitation ON
 Gen CB Closed Status
 Parallel with Grid Status
 Increase
 Decrease
 Reset Setpoint
 Remote SP Enable
 PF Enable
 Var Enable
 Manual Enable
 Open Loop Enable
 Synchronize
 VDC Enable
 Secondary Net
 Reset Alarm
 Standby
 RC Fieldbus Block
 FCB closed Status

Direction = Out

✓None
 Boost
 Field Flashing
 System OK
 Limit Active
 V/Hz Limit Active
 SP Limit Reached
 SP Minimum Reached
 SP Maximum Reached
 Operational Limit Active
 Min Ie Active
 Max Ie Active
 Min PQ Active
 Min UM Active
 Max UM Active
 Voltage Relay
 Close CB Command
 Sync Check
 SW Alarm
 Diode Alarm
 Diode Trip
 SW Alarm or Diode Trip
 MODBUS Com Alarm
 Close FCB Command
 Open FCB Command

Digital outputs

- DIO1 to 4

Parameter	Parameter INI-File	Standard value	Setting range	Set	Unit
Analog Inputs	[ANALOG INPUT]				
UM Aux: - Minimum value - Maximum value	UM Aux Minimum UM Aux Maximum	-10.0 10.0	-100.0 to +100.0 -100.0 to +100.0		% %
AI1: - Input - Low level - High level	[ANALOG INPUT 1] Input Uin 0% Uin 100%	None 0.0 10.0	Signal name -10.0 to +10.0 -10.0 to +10.0		V V
AI2: - Input - Low level - High level	[ANALOG INPUT 2] Input Uin 0% Uin 100%	None 0.0 10.0	Signal name -10.0 to +10.0 -10.0 to +10.0		V V
AI3: - Input - Low level - High level	[ANALOG INPUT 3] Input Uin 0% Uin 100%	None 0.0 10.0	Signal name -10.0 to +10.0 -10.0 to +10.0		V V
Analog Outputs	[ANALOG OUTPUT]				
AO1: - Output AO2: - Output - Low level - High level - Low level - High level	Output1 Output2 Uout1 0% Uout1 100% Uout2 0% Uout2 100%	None None 0.0 10.0 0.0 10.0	Signal name Signal name -10.0 to +10.0 -10.0 to +10.0 -10.0 to +10.0 -10.0 to +10.0		V V V V
Ie Range: Output voltage range - Low level - High level	Ie 0% Ie 100%	0.0 300.0	0 to 400.0 0 to 400.0		% %
Fbias Range: Offset signal - Low level - High level	Fbias 0% Fbias 100%	-3.0 3.0	-10.0 to +10.0 -10.0 to +10.0		Hz Hz

Signal name:

Analog inputs AI1 to 3

Direction = In

☒ None
☐ Auto Remote Setpoint
☐ PF Remote Setpoint
☐ Var Remote Setpoint
☐ Manual Remote Setpoint
☐ Open Loop Remote Setpoint
☐ UM Aux
☐ Reserved2
☐ Cooling Media Temperature
☐ Reserved4
☐ Reserved5
☐ Digital Input 9(+) & 10(-)
☐ Digital Input 11(+) & 12(-)
☐ Digital Input 13(+) & 14(-)

Direction = Out

☒ None
☐ Excitation Current
☐ PWM
☐ Fbias

Analog outputs
AO1 to 2

Parameter	Parameter INI-File	Standard value	Setting range	Set	Unit
Controller					
Auto: - Proportional gain Vp - Derivative time constant Tb - Integral time constant Ta - Reactive current influence Kq	[TUNE AUTO] Proportional Gain (Vp) Derivation Time (Tb) Integration Time (Ta) Droop (Kq)	20.0 0.20 4.00 0.0	1.0 to 200.0 0.01 to 5.00 0.01 to 20.00 -20.0 to +20.0		 s s %
PF /Var /PQ Limiter: - Proportional gain Vp - Integral time constant Ta	[TUNE PF/Var/PQ LIMITER] Proportional Gain (Vp) Integration Time (Ta)	10.0 3.00	0.1 to 200.0 0.01 to 20.00		 s
Manual/Ie Limiter: - Proportional gain Vp - Integral time constant Ta	[TUNE MANUAL/Ie LIMITER] Proportional Gain (Vp) Integration Time (Ta)	1.0 0.30	1.0 to 200.0 0.01 to 20.00		 s
Communication					
ID unit number	AVR ID	1	1 to 31		
Synchronization					
- Voltage offset	UNetoffset	0.0	-30.0 to 30.0		%
- Angle offset	Angloffset	0.0	-120.0 to 120.0		deg
- Minimum slip	Min Slip	0.00	-1.00 to 0.00		Hz
- Maximum slip	Max Slip	-0.40	-1.00 to 0.00		Hz
- Maximum voltage difference	MaxDeltaU	5.00	0 to 10.00		%
- Maximum delta angle	Max Delta Angle	10	0 to 72		deg
- Total CB closing time	Tot CB Close Time	90	0 to 630		ms
Diode Monitoring					
- Nominal exciter frequency	f Exc Nominal	50	10 to 400		Hz
- Exciter time constant	Tconst Exc	350	1 to 2000		ms
- Diode monitoring active	Active	FALSE	TRUE / FALSE		
- Alarm level	Alarm Level	5.0	2.0 to 100.0		%
- Alarm delay	Alarm Delay	10.0	2.0 to 50.0		s
- Trip level	Trip Level	20.0	10.0 to 200.0		%
- Trip delay	Trip Delay	0.3	0 to 50.0		s

Parameter	Parameter INI-File	Standard value	Setting range	Set	Unit
PSS	[PSS]				
Active, General activation	PSS_SELECT	FALSE	TRUE / FALSE		
Wash out time constants	TW1	5.0	0.0 to 30.0		s
Wash out time constants	TW2	5.0	0.0 to 30.0		s
Wash out time constants	TW3	5.0	0.0 to 30.0		s
Wash out time constants	TW4	0.0	0.0 to 30.0		s
PSS gain factor	KS1	15.0	0.1 to 50.0		p.u.
Compensation factor for calculation of integral of electric power	KS2	0.60	0.05 to 5.00		p.u.
Signal matching factor	KS3	1.00	0.05 to 5.00		p.u.
Lead time constants of conditioning network	T1	0.10	0.02 to 6.00		s
Lead time constants of conditioning network	T3	0.18	0.02 to 6.00		s
Lead time constants of conditioning network	T10	0.26	0.02 to 6.00		s
Lag time constants of conditioning network	T2	0.02	0.02 to 6.00		s
Lag time constants of conditioning network	T4	0.02	0.02 to 6.00		s
Lag time constants of conditioning network	T11	0.02	0.02 to 6.00		s
Time constant for integral of electrical power	T7	5.00	0.00 to 30.00		s
Ramp tracking filter time constant	T8	0.00	0.00 to 2.50		s
Ramp tracking filter time constant	T9	0.10	0.00 to 2.50		s
Ramp tracking filter degree	M	5	2 to 5		
Ramp tracking filter degree	N	1	-1 to 4		
Upper limit of stabilizing signal	PSS max	10.0	0.0 to 25.0		%
Lower limit of stabilizing signal	PSS min	-10.0	0.0 to -25.0		%
Activation level for PSS functionality	P min	5.0	0.0 to 50.0		%
IM Limiter	[IM Limiter]				
- Limiter high level	IMlim_Active	FALSE	TRUE / FALSE		
- Time Multiplier K	IMlim_K	1.00	0.01 to 100.00		
- Maximum	IMlim_Im_th	100.0	0.0 to 100.0		%
TEMP INFLUENCE	[TEMP INFLUENCE]				
- T1 at Max IM	IMlim_T1	50.0	-100.0 to 100.0		°
- Machine current level	IMlim_Im_th2	75.0	0.0 to 100.0		%
- T2 at IM th2	IMlim_T2	60.0	-100.0 to 100.0		°
- Machine current level	IMlim_Im_th3	50.0	0.0 to 100.0		%
- T3 at IM th3	IMlim_T3	70.0	-100.0 to 100.0		°
- T4 at IM = 0 A	IMlim_T4	80.0	-100.0 to 100.0		°
- T1 at Max Continuous Ie	IElim_T1	50.0	-100.0 to 100.0		°
- Excitation current level	IElim_Ie_th2	75.0	0.0 to 100.0		%
- T2 at Ie th2	IElim_T2	60.0	-100.0 to 100.0		°
- Excitation current level	IElim_Ie_th3	50.0	0.0 to 100.0		%
- T3 at Ie th3	IElim_T3	70.0	-100.0 to 100.0		°
- T4 at Ie = 0 A	IElim_T4	80.0	-100.0 to 100.0		°

Parameter	Parameter INI-File	Standard value	Setting range	Set	Unit
MODBUS	[MODBUS]				
Slave ID	MB_Slave	247	1 to 247		
Protocol	MB_Protocol	0	0 to 1		
Bit rate	MB_Bitrate	7	3 to 7 and 10 to 12		
Character framing	MB_ChrFrame	0	0 to 3		
Keep-alive time	MB_KeepAliveTime	0	0 to 120		s
Keep-alive actions	MB_KeepAliveAction	0	0 to 3		
MODBUS active	MB_Enable	FALSE	TRUE / FALSE		
Answer delay	MB_Delay	10	10 to 500 (step of 10)		ms

Value	Protocol
0	☒ RTU
1	☐ TCP

Value	Bit rate
0	RESERVED
1	RESERVED
2	RESERVED
3	115200
4	57600
5	38400
6	28800
7	☒ 19200
8	RESERVED
9	
10	9600
11	4800
12	2400

Value	Character framing
0	☒ Even
1	☐ Odd
2	☐ none
3	☐ none

Value	Keep-alive action
0	☒ none
1	☐ Set SW alarm
2	☐ Set fieldbus block
3	☐ Clear fieldbus request

Values have to be saved in the unit.

- Display: Start menu **MAIN, Write EEPROM, OK, OK**
- CMT 1000:
 - Unit: **Menu bar CMT 1000 \ File \ Write Parameters to EEPROM**
 - Harddisk: **Menu bar CMT 1000 \ File \ Save Parameter File**

REVISION

Rev. ind.	Page (P) Chapt. (C)	Description (or number of revision)	Date Dept./Init.
A	33	Cross sections	13.12.2004/Mo
B	78	According to UL specifications Includes the Power System Stabilizer parts	03.06.2005/Mo
C		Synchronizing, Angle and Voltage Offset IM Limiter Temperature depending, Ie and IM MODBUS FCB Control CMT 1000, Panel	20.01.2006/Mo